Learning Latent Fields in Multiobject Tracking: Supplementary Material

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This manuscript provides additional derivations and results for the paper [1]. Equations, figures, and tables labeled with a section prefix (e.g., (I.1), Fig. I.1, Table I.1) refer to this supplementary material; all other references refer to [1].

The remainder of this manuscript is organized as follows. Section I derives the evidence lower bound (ELBO) used to learn the latent field during runtime without labeled data and summarizes the processing at one time step. Section II derives the forward Bayesian inference, including the sum-product message passing, the Gaussian approximation of the marginal likelihoods, and their Monte Carlo implementation with weighted particles. Section III develops a low-rank and Kronecker-structured formulation that makes the inference computationally feasible for high-dimensional raw measurements and analyzes its storage and computational complexity. Section IV presents additional results for the simulated multisensor radar scenario. Section V provides additional results for the active radio sensing experiment, including the posterior PDF at one time step, the evolution of the learned signal-power PDF through the flow transformations, and the sensitivity to birth initialization. Section VI compares the learned latent field with KRAKEN predictions for the passive acoustic sensing experiment. Section VII presents the ablation study of the signal-power prior in a controlled acoustic environment.

I. Derivations of the Variational Learning for the Latent Field

This section derives the learning objective in our proposed method. We first review the system model.

A. System Model and Notation

We consider a single raw measurement vector $\mathbf{z}_k \in \mathbb{C}^C$ for a given time step k, N_k potential objects, and a scalar random variable $\eta_k \in \mathbb{R}_{>0}$ to denote the noise variance. For each object $n \in \{1, \ldots, N_k\}$ we define

$$\mathbf{x}_{k,n} \in \mathbb{R}^d, \qquad \boldsymbol{\phi}_{k,n} = (r_{k,n},\, \gamma_{k,n}),$$

where $\mathbf{x}_{k,n}$ is a random variable denoting the kinematic state (e.g., $d = 4$ for 2-D position and velocity), $r_{k,n} \in \{0, 1\}$ is the binary existence variable, and $\gamma_{k,n} > 0$ is a random variable denoting the signal power of object n, i.e., the conditional mean squared magnitude of its complex amplitude.

1) Likelihood: Given the complex amplitude $\rho_{k,n} \sim \mathcal{CN}(0, \gamma_{k,n})$ and noise $\mathbf{w}_k \sim \mathcal{CN}(\mathbf{0}, \eta_k \mathbf{I}_C)$, the measurement at time k is taken as

$$\mathbf{z}_k = \sum_{n=1}^{N_k} r_{k,n}\, \rho_{k,n}\, \mathbf{a}(\mathbf{x}_{k,n}) + \mathbf{w}_k. \tag{I.1}$$

Conditional on the existence variables $\mathbf{r}_k = r_{k,1:N_k}$, the kinematic states $\mathbf{x}_k = \mathbf{x}_{k,1:N_k}$, the signal powers $\boldsymbol{\gamma}_k = \gamma_{k,1:N_k}$, and η_k, the amplitudes $\rho_{k,n}$ are mutually independent and independent of the noise $\mathbf{w}_k$, and all of them are proper complex Gaussian. Marginalizing the amplitudes yields a zero-mean complex Gaussian likelihood:

$$f(\mathbf{z}_k \mid \mathbf{r}_k, \mathbf{x}_k, \boldsymbol{\gamma}_k, \eta_k) = \mathcal{CN}\big(\mathbf{z}_k;\, \mathbf{0},\, \boldsymbol{\Sigma}_k\big), \tag{I.2}$$

$$\boldsymbol{\Sigma}_k = \sum_{n=1}^{N_k} r_{k,n}\gamma_{k,n}\, \mathbf{a}(\mathbf{x}_{k,n})\mathbf{a}(\mathbf{x}_{k,n})^{\mathrm{H}} + \eta_k\, \mathbf{I}_C, \tag{I.3}$$

where $\mathbf{a}(\cdot) \in \mathbb{C}^C$ denotes the physics-based array manifold, which is in general a nonlinear function of the kinematic state.

2) Prior PDF: At each time step k, every random variable has a prior density as

$$f(\mathbf{x}_{k,n}), \qquad f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n}), \qquad f(\eta_k), \tag{I.4}$$

where the prior $f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})$ is denoted as

$$f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n}) = \begin{cases} f(r_{k,n}=1)\, f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n}=1; \mathbf{c}_{k,n}), & r_{k,n}=1, \\ f(r_{k,n}=0)\, f_{\mathrm{D}}(\gamma_{k,n}), & r_{k,n}=0, \end{cases} \tag{I.5}$$

where $f_{\mathrm{D}}(\gamma_{k,n})$ is a *dummy* PDF assigned when $r_{k,n}=0$, since the measurement model does not depend on $\gamma_{k,n}$ in this case and it can be assigned an arbitrary reference PDF without affecting inference. The learnable component is the conditional prior $f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n}=1; \mathbf{c}_{k,n})$, which equals the signal-power PDF $f_{\boldsymbol{\theta}}(\gamma_{k,n}; \mathbf{c}_{k,n})$ of the latent field (cf. (10) of the main paper). The latent field is the family of context-dependent signal-power PDFs introduced in Section III-C of the main paper, and its parameters $\boldsymbol{\theta}$ are shared across objects and time. The context $\mathbf{c}_{k,n}$ of object n at time step k comprises its position and possibly known sensing parameters and signal features. It is treated as a deterministic quantity, not a random variable.

The predicted PDFs $f(r_{k,n})$, $f(\mathbf{x}_{k,n})$, and $f(\eta_k)$ are not learned. As is common, they are obtained from first-order Markov models with the transition PDFs $f(r_{k,n} \mid r_{k-1,n})$, $f(\mathbf{x}_{k,n} \mid \mathbf{x}_{k-1,n})$, and $f(\eta_k \mid \eta_{k-1})$, as described in Section IV of the main paper. For example,

$$f(\mathbf{x}_{k,n}) = \int f(\mathbf{x}_{k,n} \mid \mathbf{x}_{k-1,n})\, q(\mathbf{x}_{k-1,n})\, \mathrm{d}\mathbf{x}_{k-1,n}$$

where $q(\mathbf{x}_{k-1,n})$ is the marginal posterior PDF of time step $k-1$. However, the variational framework developed below does not depend on these specific choices. The concrete form of $f_{\boldsymbol{\theta}}(\cdot)$ – either a nonparametric or parametric statistical family – is specified below.

3) Posterior PDF: Let $\mathbf{h}_k = \big(\mathbf{x}_{k,1:N_k}, \boldsymbol{\phi}_{k,1:N_k}, \eta_k\big)$ collect all hidden variables at time k. We adopt a sequential filtering perspective, where the predictive prior before observing $\mathbf{z}_k$ is $f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_{1:k-1})$. For notational simplicity, we denote this predictive prior by $f_{\boldsymbol{\theta}}(\mathbf{h}_k)$. The joint density is then written as

$$f_{\boldsymbol{\theta}}(\mathbf{z}_k, \mathbf{h}_k) = f(\mathbf{z}_k \mid \mathbf{h}_k)\, f_{\boldsymbol{\theta}}(\mathbf{h}_k), \tag{I.6}$$

where $f(\mathbf{z}_k \mid \mathbf{h}_k) = \mathcal{CN}\big(\mathbf{z}_k; \mathbf{0}, \boldsymbol{\Sigma}_k\big)$ and $f_{\boldsymbol{\theta}}(\mathbf{h}_k) = f(\eta_k) \prod_n f(\mathbf{x}_{k,n}) f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})$. The dependence on previous measurements is absorbed into the predictive prior through posterior propagation. The evidence at time k is obtained as

$$f_{\boldsymbol{\theta}}(\mathbf{z}_k) = \int f(\mathbf{z}_k \mid \mathbf{h}_k) f_{\boldsymbol{\theta}}(\mathbf{h}_k) d\mathbf{h}_k, \tag{I.7}$$

where $f_{\boldsymbol{\theta}}(\mathbf{z}_k)$ abbreviates $f_{\boldsymbol{\theta}}(\mathbf{z}_k \mid \mathbf{z}_{1:k-1})$. Therefore, the posterior PDF at time step k is given by Bayes' rule:

$$f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_k) = \frac{f_{\boldsymbol{\theta}}(\mathbf{z}_k, \mathbf{h}_k)}{f_{\boldsymbol{\theta}}(\mathbf{z}_k)}. \tag{I.8}$$

Direct computation of (I.8) is generally intractable. We therefore approximate the posterior PDF $f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_k)$ by a PDF $q(\mathbf{h}_k)$. This PDF serves as a formal auxiliary density for the derivation; belief propagation (BP) computes only its marginal posterior PDFs, which are represented by weighted particles. The resulting marginal posterior PDFs are obtained by marginalizing $q(\mathbf{h}_k)$:

$$q(\mathbf{x}_{k,n}) = \int q(\mathbf{h}_k)\, d\boldsymbol{\phi}_{k,1:N_k}\, d\eta_k \prod_{n' \neq n} d\mathbf{x}_{k,n'} \propto f(\mathbf{x}_{k,n})\, \ell_{\mathbf{x},n}(\mathbf{x}_{k,n}), \tag{I.9}$$

$$q(\boldsymbol{\phi}_{k,n}) = \int q(\mathbf{h}_k)\, d\mathbf{x}_{k,1:N_k}\, d\eta_k \prod_{n' \neq n} d\boldsymbol{\phi}_{k,n'} \propto f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\, \ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_{k,n}), \tag{I.10}$$

$$q(\eta_k) = \int q(\mathbf{h}_k)\, d\mathbf{x}_{k,1:N_k}\, d\boldsymbol{\phi}_{k,1:N_k} \propto f(\eta_k)\, \ell_{\eta}(\eta_k), \tag{I.11}$$

where each marginal is obtained by integrating out all hidden variables in $\mathbf{h}_k$ except the one of interest; for the binary existence variables $r_{k,n'}$ contained in $\boldsymbol{\phi}_{k,n'}$, the integration is understood as a summation over $\{0,1\}$. Here $\ell_{\mathbf{x},n}(\mathbf{x}_{k,n})$, $\ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_{k,n})$, and $\ell_{\eta}(\eta_k)$ denote the marginal likelihood terms associated with the kinematic state of object n, its existence and signal power, and

the noise variance at time k, respectively. The proportionality relations hold exactly for the factor graph of a single measurement block, which is a tree (cf. Section II-B). The Gaussian approximation in Section II-C and, for multiple measurement blocks, the loops of the factor graph make them approximate, i.e., the marginal posterior PDFs are then approximated by the beliefs of BP.

B. Variational Inference

We derive the learning objective from the standard evidence decomposition. At time step k, for any fixed distribution $q(\mathbf{h}_k)$, the one-step evidence satisfies

$$\log f_{\boldsymbol{\theta}}(\mathbf{z}_k) = \mathcal{L}_k(\boldsymbol{\theta}) + \mathrm{KL}(q(\mathbf{h}_k) \,\|\, f_{\boldsymbol{\theta}}(\mathbf{h}_k \mid \mathbf{z}_k)) \,, \tag{I.12}$$

where the corresponding ELBO is

$$\mathcal{L}_k(\boldsymbol{\theta}) = \mathbb{E}_{q(\mathbf{h}_k)} \left[\log f_{\boldsymbol{\theta}}(\mathbf{z}_k, \mathbf{h}_k) - \log q(\mathbf{h}_k) \right]. \tag{I.13}$$

Since the KL divergence is nonnegative, $\mathcal{L}_k(\boldsymbol{\theta})$ is a lower bound on the evidence $\log f_{\boldsymbol{\theta}}(\mathbf{z}_k)$. Using the factorization in (I.6), the ELBO can be written as

$$\begin{aligned} \mathcal{L}_k(\boldsymbol{\theta}) = &\underbrace{\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f(\mathbf{z}_k \mid \mathbf{h}_k)\big]}_{\text{likelihood}} + \underbrace{\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k})\big]}_{\text{signal-power prior}} \\ &+ \underbrace{\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f(\mathbf{x}_{k,1:N_k}) + \log f(\eta_k)\big]}_{\text{other priors}} - \underbrace{\mathbb{E}_{q(\mathbf{h}_k)}\big[\log q(\mathbf{h}_k)\big]}_{\text{entropy}}. \end{aligned} \tag{I.14}$$

When $q(\mathbf{h}_k)$ is fixed during the flow update, the likelihood term, the kinematic prior, the noise prior, and the entropy term are independent of $\boldsymbol{\theta}$. Therefore, the $\boldsymbol{\theta}$-dependent part of $\mathcal{L}_k(\theta)$ comes only from the prior $f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k})$. Thus, we only need to evaluate the expected log-prior $\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k})\big]$, which we expand step by step in the following.

First, the prior of the existence and signal-power states factorizes across objects, i.e., $f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k}) = \prod_{n=1}^{N_k} f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})$. The logarithm turns this product into a sum, and by the linearity of expectation we obtain

$$\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k})\big] = \mathbb{E}_{q(\mathbf{h}_k)}\left[\sum_{n=1}^{N_k} \log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\right] = \sum_{n=1}^{N_k} \mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\big]. \tag{I.15}$$

Second, each summand in (I.15) depends on $\mathbf{h}_k$ only through $\boldsymbol{\phi}_{k,n}$. Integrating out all remaining hidden variables, i.e., $\mathbf{x}_{k,1:N_k}$, $\boldsymbol{\phi}_{k,n'}$ with $n' \neq n$, and η_k, yields the marginal posterior PDF $q(\boldsymbol{\phi}_{k,n})$, so that

$$\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\big] = \int q(\mathbf{h}_k)\, \log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\, d\mathbf{h}_k = \int q(\boldsymbol{\phi}_{k,n})\, \log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\, d\boldsymbol{\phi}_{k,n} = \mathbb{E}_{q(\boldsymbol{\phi}_{k,n})}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\big]. \tag{I.16}$$

Third, since $\boldsymbol{\phi}_{k,n} = (r_{k,n}, \gamma_{k,n})$ consists of a binary existence variable and a continuous signal power, the expectation in (I.16) is a sum over $r \in \{0,1\}$ and an integral over $\gamma_{k,n}$. Factorizing the marginal posterior as $q(r_{k,n} = r, \gamma_{k,n}) = q(r_{k,n} = r)\, q(\gamma_{k,n} \mid r_{k,n} = r)$, we have

$$\begin{aligned} \mathbb{E}_{q(\boldsymbol{\phi}_{k,n})}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,n})\big] &= \sum_{r\in\{0,1\}} \int q(r_{k,n} = r, \gamma_{k,n})\, \log f_{\boldsymbol{\theta}}(r_{k,n} = r, \gamma_{k,n})\, d\gamma_{k,n} \\ &= \sum_{r\in\{0,1\}} q(r_{k,n} = r) \int q(\gamma_{k,n} \mid r_{k,n} = r)\, \log f_{\boldsymbol{\theta}}(r_{k,n} = r, \gamma_{k,n})\, d\gamma_{k,n} \\ &= \sum_{r\in\{0,1\}} q(r_{k,n} = r)\, \mathbb{E}_{q(\gamma_{k,n}|r_{k,n}=r)}\big[\log f_{\boldsymbol{\theta}}(r_{k,n} = r, \gamma_{k,n})\big]. \end{aligned} \tag{I.17}$$

Fourth, according to the prior in (I.5), the logarithm of the prior evaluated for the two existence hypotheses reads

$$\log f_{\boldsymbol{\theta}}(r_{k,n} = 1, \gamma_{k,n}) = \log f(r_{k,n} = 1) + \log f_{\boldsymbol{\theta}}\big(\gamma_{k,n} \mid r_{k,n} = 1; \mathbf{c}_{k,n}\big), \tag{I.18}$$

$$\log f_{\boldsymbol{\theta}}(r_{k,n} = 0, \gamma_{k,n}) = \log f(r_{k,n} = 0) + \log f_{\mathrm{D}}(\gamma_{k,n}). \tag{I.19}$$

Finally, substituting (I.16), (I.17), (I.18), and (I.19) into (I.15), the expected log-prior is obtained as

$$\begin{aligned}
&\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k})\big] \\
&= \sum_{n=1}^{N_k} \Big[q(r_{k,n}=1)\mathbb{E}_{q(\gamma_{k,n}|r_{k,n}=1)}\Big[\log f(r_{k,n}=1) + \log f_{\boldsymbol{\theta}}\big(\gamma_{k,n} \mid r_{k,n}=1; \mathbf{c}_{k,n}\big)\Big] \\
&\qquad + q(r_{k,n}=0)\mathbb{E}_{q(\gamma_{k,n}|r_{k,n}=0)}\Big[\log f(r_{k,n}=0) + \log f_{\mathrm{D}}(\gamma_{k,n})\Big]\Big].
\end{aligned} \tag{I.20}$$

Since $\log f(r_{k,n}=r)$ does not depend on $\gamma_{k,n}$ and $\int q(\gamma_{k,n} \mid r_{k,n}=r)\, d\gamma_{k,n} = 1$, the existence terms can be taken out of the expectations. Hence, (I.20) can be separated into a $\boldsymbol{\theta}$-dependent part and a $\boldsymbol{\theta}$-independent part, i.e.,

$$\begin{aligned}
\mathbb{E}_{q(\mathbf{h}_k)}\big[\log f_{\boldsymbol{\theta}}(\boldsymbol{\phi}_{k,1:N_k})\big] &= \sum_{n=1}^{N_k} q(r_{k,n}=1)\, \mathbb{E}_{q(\gamma_{k,n}|r_{k,n}=1)}\Big[\log f_{\boldsymbol{\theta}}\big(\gamma_{k,n} \mid r_{k,n}=1; \mathbf{c}_{k,n}\big)\Big] \\
&\quad + \underbrace{\sum_{n=1}^{N_k}\Big(\sum_{r\in\{0,1\}} q(r_{k,n}=r)\log f(r_{k,n}=r) + q(r_{k,n}=0)\, \mathbb{E}_{q(\gamma_{k,n}|r_{k,n}=0)}\big[\log f_{\mathrm{D}}(\gamma_{k,n})\big]\Big)}_{\text{independent of } \boldsymbol{\theta}}.
\end{aligned} \tag{I.21}$$

Here $q(r_{k,n}=1)$ and $q(\gamma_{k,n} \mid r_{k,n}=1)$ are obtained from the marginal posterior PDF $q(\boldsymbol{\phi}_{k,n})$. Since $f(r_{k,n})$ and the dummy density $f_{\mathrm{D}}(\gamma_{k,n})$ are independent of $\boldsymbol{\theta}$, the last term in (I.21) is a constant. Therefore, maximizing the ELBO with respect to $\boldsymbol{\theta}$ is equivalent to maximizing its $\boldsymbol{\theta}$-dependent part:

$$\mathcal{J}_k(\boldsymbol{\theta}) = \sum_n q(r_{k,n}=1)\mathbb{E}_{q(\gamma_{k,n}|r_{k,n}=1)}\left[\log f_{\boldsymbol{\theta}}\big(\gamma_{k,n} \mid r_{k,n}=1; \mathbf{c}_{k,n}\big)\right]. \tag{I.22}$$

At time step k, the accumulated objective is given by

$$\mathcal{J}_{1:k}(\boldsymbol{\theta}) = \sum_{k'=1}^{k} \mathcal{J}_{k'}(\boldsymbol{\theta}) \tag{I.23}$$

We now describe how the expectation in $\mathcal{J}_k(\boldsymbol{\theta})$ is evaluated in practice. Since the posterior q is represented nonparametrically via Monte Carlo method (Section II-D), the conditional posterior of $\gamma_{k,n}$ given $r_{k,n}=1$ is represented by a weighted particle set:

$$q(\gamma_{k,n} \mid r_{k,n}=1) \approx \left\{\big(\omega_{\gamma,k,n,p}, \gamma_{k,n,p}\big)\right\}_{p=1}^{P}, \tag{I.24}$$

where the weights satisfy $\sum_{p=1}^{P} \omega_{\gamma,k,n,p} = 1$. Substituting (I.24) into (I.22) gives the Monte Carlo objective as:

$$\mathcal{J}_{1:k}(\boldsymbol{\theta}) = \sum_{k'=1}^{k}\sum_{n} q(r_{k',n}=1)\sum_{p=1}^{P} \omega_{\gamma,k',n,p} \log f_{\boldsymbol{\theta}}\big(\gamma_{k',n,p} \mid r_{k',n}=1; \mathbf{c}_{k',n}\big). \tag{I.25}$$

During the parameter update, the inferred particles and their weights are treated as fixed outputs of the forward inference. Therefore, the gradient with respect to $\boldsymbol{\theta}$ acts only on the flow density:

$$\nabla_{\boldsymbol{\theta}}\mathcal{J}_{1:k}(\boldsymbol{\theta}) = \sum_{k'=1}^{k}\sum_{n} q(r_{k',n}=1)\sum_{p=1}^{P} \omega_{\gamma,k',n,p} \nabla_{\boldsymbol{\theta}} \log f_{\boldsymbol{\theta}}\big(\gamma_{k',n,p} \mid r_{k',n}=1; \mathbf{c}_{k',n}\big). \tag{I.26}$$

The posterior power particles are samples from the prior PDF obtained with the parameters before the update, reweighted by the likelihood. Hence, maximizing (I.25) amounts to a weighted maximum-likelihood fit whose support is restricted to that of the preceding prior, and its accuracy depends on the effective sample size $1/\sum_{p=1}^{P} \omega_{\gamma,k',n,p}^2$ of the power particles.

In the following sections, we discuss several choices of this conditional PDF.

C. Nonparametric Prior PDF via Conditional Normalizing Flow

We model the conditional prior $f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n}=1; \mathbf{c}_{k,n})$, i.e., the signal-power PDF of the latent field, using a conditional normalizing flow as in Section III-D of the main paper, which provides a flexible family of PDFs over $\gamma_{k,n}$. Since $\gamma_{k,n} > 0$, we first map to an unconstrained domain:

$$\gamma_{k,n} = G(\mu_{k,n}), \qquad \mu_{k,n} \in \mathbb{R}, \tag{I.27}$$

where $G(\cdot) : \mathbb{R} \to \mathbb{R}_{>0}$ is a smooth bijection. For notational simplicity, we suppress the indices (k, n) in the following derivation. We then model the conditional PDF $f_{\boldsymbol{\theta}}(\mu; \mathbf{c})$ using a conditional normalizing flow. Let $f_0(\varepsilon)$ be a base PDF on $\mathbb{R}$ (e.g., $\varepsilon \sim \mathcal{N}(0, 1)$). For a given context $\mathbf{c}$ as input, define a composition of Q invertible, monotone transformations:

$$y_0 = \varepsilon, \qquad y_\ell = F_{\theta_\ell}(y_{\ell-1}; \mathbf{c}), \quad \ell = 1, \ldots, Q, \tag{I.28}$$

where each $F_{\theta_\ell}(\cdot; \mathbf{c}) : \mathbb{R} \to \mathbb{R}$ is a differentiable bijection parameterized by θ_ℓ and conditioned on $\mathbf{c}$. In particular, $F_{\theta_\ell}(\cdot; \mathbf{c})$ is a nonlinear, strictly monotone function whose coefficients are computed from the context $\mathbf{c}$ by an encoder, i.e., a neural network with parameters θ_ℓ. The output is $\mu = y_Q = T_{\boldsymbol{\theta}}(\varepsilon; \mathbf{c})$, with $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_Q)$. By the change-of-variables formula, the log-PDF of μ given $\mathbf{c}$ is

$$\log f_{\boldsymbol{\theta}}(\mu; \mathbf{c}) = \log f_0(\varepsilon) - \sum_{\ell=1}^{Q} \log \left| \frac{\partial F_{\theta_\ell}(y_{\ell-1}; \mathbf{c})}{\partial y_{\ell-1}} \right|, \tag{I.29}$$

where ε and $\{y_\ell\}$ are recovered by inverting the flow for the given $(\mu, \mathbf{c})$. Transforming back to the original signal-power variable via $\gamma = G(\mu)$, we obtain

$$\log f_{\boldsymbol{\theta}}(\gamma \mid r=1; \mathbf{c}) = \log f_{\boldsymbol{\theta}}(\mu; \mathbf{c}) - \log \left| \frac{dG(\mu)}{d\mu} \right|. \tag{I.30}$$

Substituting (I.30) into the Monte Carlo objective (I.25) yields the objective that is maximized to train the flow parameters $\boldsymbol{\theta}$. Before the first time step, $\boldsymbol{\theta}$ is initialized such that the flow yields the uniform PDF on $(0, \gamma_{\max}]$ for all contexts.

1) Training and Inference Pipeline: The training and inference procedures are coupled during multiobject tracking. The complete procedure for a given context $\mathbf{c}$ consists of four steps:

1) **Prior sampling.** For each power particle index $p = 1, \ldots, P$:
   - Draw a base sample $\varepsilon_p \sim f_0(\varepsilon)$.
   - Propagate through the flow: $\mu_p = T_{\boldsymbol{\theta}}(\varepsilon_p; \mathbf{c})$.
   - Transform to signal-power space: $\gamma_p = G(\mu_p) > 0$.

   This yields prior particles $\{\gamma_p\}_{p=1}^{P} \sim f_{\boldsymbol{\theta}}(\gamma \mid r=1; \mathbf{c})$ with uniform weights.
2) **Posterior reweighting.**
   - After the new measurement is received, each prior particle γ_p is reweighted by importance sampling, which yields the posterior weights $\omega_{\gamma,p}$.
   - The resulting particle set $q(\gamma \mid r=1) \approx \{(\omega_{\gamma,p}, \gamma_p)\}_{p=1}^{P}$ approximates the marginal posterior.
3) **Density evaluation.** For each posterior particle γ_p:
   - Invert the transformation: $\mu_p = G^{-1}(\gamma_p)$.
   - Invert the flow to recover the base sample ε_p and the intermediate values $\{y_\ell\}$.
   - Compute $\log f_{\boldsymbol{\theta}}(\gamma_p \mid r=1; \mathbf{c})$ via (I.30) by accumulating the Jacobian terms.
4) **Parameter update.**
   - Evaluate the gradient $\nabla_{\boldsymbol{\theta}} \mathcal{J}_{1:k}(\boldsymbol{\theta})$ approximately using the posterior weights ω_p and the log-PDFs from step 3, which are buffered across time steps and scenes. A window is applied if available memory becomes limited.

D. Parametric Prior PDF

When a parametric family is considered, for example, we model the conditional prior as a Gamma PDF

$$f_{\boldsymbol{\theta}}(\gamma_{k,n} \mid r_{k,n}=1;\, \mathbf{c}_{k,n}) = \mathrm{Gamma}(\gamma_{k,n}\,;\, \alpha_{\boldsymbol{\theta}}(\mathbf{c}_{k,n}),\, \beta_{\boldsymbol{\theta}}(\mathbf{c}_{k,n}))\,, \tag{I.31}$$

where the shape $\alpha_{\boldsymbol{\theta}}(\cdot) > 0$ and rate $\beta_{\boldsymbol{\theta}}(\cdot) > 0$ are produced by a neural network using the context $\mathbf{c}_{k,n}$ as input. Substituting into the Monte Carlo objective (I.25) and expanding the Gamma log-PDF gives

$$\begin{aligned}\mathcal{J}_{1:k}(\boldsymbol{\theta}) &= \sum_{k'=1}^{k} \sum_{n}\; q(r_{k',n}=1) \sum_{p} \omega_{\gamma,k',n,p}\, \log \mathrm{Gamma}(\gamma_{k',n,p};\, \alpha_{\boldsymbol{\theta}}(\mathbf{c}_{k',n}),\, \beta_{\boldsymbol{\theta}}(\mathbf{c}_{k',n})) \\ &= \sum_{k'=1}^{k} \sum_{n}\; q(r_{k',n}=1) \Big[\alpha_{\boldsymbol{\theta}}(\mathbf{c}) \log \beta_{\boldsymbol{\theta}}(\mathbf{c}) - \log \Gamma\big(\alpha_{\boldsymbol{\theta}}(\mathbf{c})\big) + \big(\alpha_{\boldsymbol{\theta}}(\mathbf{c}) - 1\big)\, \overline{\log \gamma}_{k',n} - \beta_{\boldsymbol{\theta}}(\mathbf{c})\, \hat{\gamma}_{k',n}\Big],\end{aligned} \tag{I.32}$$

where $\mathbf{c} = \mathbf{c}_{k',n}$, and the sufficient statistics are

$$\overline{\log \gamma}_{k',n} = \sum_{p} \omega_{\gamma,k',n,p} \log \gamma_{k',n,p}, \qquad \hat{\gamma}_{k',n} = \sum_{p} \omega_{\gamma,k',n,p}\, \gamma_{k',n,p}. \tag{I.33}$$

E. Processing at One Time Step

Algorithm 1 summarizes the processing at time step k, which combines the prediction step described in Section IV of the main paper, the Bayesian inference in Section II, its efficient implementation in Section III, and the parameter update derived in this section.

Algorithm 1: Processing at time step k

Input: Particle representations of $q(\mathbf{x}_{k-1,n})$ and $q(\eta_{k-1})$, existence probabilities $q(r_{k-1,n}=1)$, flow parameters $\boldsymbol{\theta}$, and measurement $\mathbf{z}_k$

Output: State estimates, particle representations of the marginal posterior PDFs at time step k, and updated parameters $\boldsymbol{\theta}$

Prediction: propagate the particles of the kinematic states of legacy potential objects and of the noise variance through the transition PDFs, compute $f(r_{k,n}=1) = p_{\mathrm{s}}\, q(r_{k-1,n}=1)$, and introduce new potential objects with birth probabilities $p_{\mathrm{B},n}$ and birth PDFs $f_{\mathrm{B},n}(\mathbf{x}_{k,n})$;

Context: compute the contexts $\mathbf{c}_{k,n}$ of all potential objects;

Prior sampling: draw the power particles $\gamma_{k,n,p}$ from $f_{\boldsymbol{\theta}}(\gamma; \mathbf{c}_{k,n})$ using the flow (Section I-C1);

Bayesian inference: compute the marginal likelihoods with the formulation in Section III, and update the particle weights (II.27) and the existence probabilities (II.28); for multiple measurement blocks, iterate the messages (Section II-D);

Estimation: declare the potential objects with $q(r_{k,n}=1) > T_{\mathrm{dec}}$ to exist and compute the MMSE estimates of their kinematic states;

Learning: store $q(r_{k,n}=1)$, the weighted power particles, and $\mathbf{c}_{k,n}$, and update $\boldsymbol{\theta}$ by a few gradient steps that increase (I.25);

Pruning and resampling: remove the potential objects with $q(r_{k,n}=1) < T_{\mathrm{pru}}$ and resample the particles of the kinematic states and of the noise variance;

II. Derivations of Forward Bayesian Inference

This section details the forward Bayesian inference of our proposed method. The key derivations proceed in three main steps:

1) We use the sum-product algorithm to compute the marginal likelihood messages in a factor graph (Section II-B).
2) We show that these marginal likelihoods are zero-mean continuous Gaussian mixtures. To ensure analytical tractability, we approximate each likelihood as a single zero-mean Gaussian PDF via second-order moment matching (Section II-C).
3) Because the integrals required to compute the marginal likelihoods remain intractable for nonparametric priors, we employ Monte Carlo methods to complete the Bayesian inference procedure, where all prior and posterior PDFs are represented by weighted particles (Section II-D).

A. Model and Notation

We consider a single time step with a single raw measurement vector $\mathbf{z} \in \mathbb{C}^C$, N potential objects, and a scalar noise parameter $\eta \in \mathbb{R}_{>0}$. For each object $n \in \{1, \ldots, N\}$ we define

$$\mathbf{x}_n \in \mathbb{R}^d, \qquad \boldsymbol{\phi}_n = (r_n,\, \gamma_n),$$

where $\mathbf{x}_n$ is the kinematic state (e.g., $d = 4$ for 2-D position and velocity), $r_n \in \{0, 1\}$ is the existence variable of object n, and $\gamma_n > 0$ is the signal power of object n. The time index k is omitted in this section.

The likelihood is

$$f_{\mathbf{z}} \equiv f(\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta) = \mathcal{CN}\big(\mathbf{z}; \mathbf{0}, \boldsymbol{\Sigma}\big). \tag{II.1}$$

with covariance

$$\boldsymbol{\Sigma} = \eta \mathbf{I}_C + \sum_{n=1}^{N} r_n \gamma_n \, \mathbf{a}(\mathbf{x}_n)\mathbf{a}(\mathbf{x}_n)^{\mathrm{H}}. \tag{II.2}$$

Here, $\mathbf{a}(\mathbf{x}_n)$ is the array response of object n. We use the following simplified notations to denote the prior PDFs

$$f_{\mathbf{x},n} \equiv f(\mathbf{x}_n), \qquad f_{\boldsymbol{\phi},n} \equiv f(\boldsymbol{\phi}_n), \qquad f_{\eta} \equiv f(\eta), \tag{II.3}$$

The marginal posterior PDFs are computed as

$$q(\mathbf{x}_n \mid \mathbf{z}) \propto f(\mathbf{x}_n) \cdot \ell_{\mathbf{x},n}(\mathbf{x}_n) \tag{II.4}$$

$$q(\boldsymbol{\phi}_n \mid \mathbf{z}) \propto f(\boldsymbol{\phi}_n) \cdot \ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_n) \tag{II.5}$$

$$q(\eta \mid \mathbf{z}) \propto f(\eta) \cdot \ell_{\eta}(\eta) \tag{II.6}$$

where $\ell_{\mathbf{x},n}(\mathbf{x}_n)$, $\ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_n)$, and $\ell_{\eta}(\eta)$ denote the marginal likelihoods for the kinematic state of object n, for its existence variable and signal power, and for the noise variance, respectively.

B. Likelihood Computation

The marginal likelihoods in (II.4)–(II.6) can be computed by applying the sum-product algorithm [3], i.e., BP, to the factor graph shown in Fig. II.1. Because the measurement is a coherent superposition of the contributions of all objects, the marginal likelihood for one object is obtained by marginalizing the contributions of all other objects. Table II.1 summarizes the key quantities used in Sections II-A–II-D.

From the factor graph, the marginal likelihood to the random variable $\boldsymbol{\phi}_n$ can be computed by the sum-product algorithm:

$$\begin{aligned}\ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_n) = &\int \cdots \int f(\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta)\, f(\eta)\, d\eta \\ &\times \prod_{m \neq n} f(\boldsymbol{\phi}_m)\, d\boldsymbol{\phi}_m \prod_{m=1}^{N} f(\mathbf{x}_m)\, d\mathbf{x}_m,\end{aligned} \tag{II.7}$$

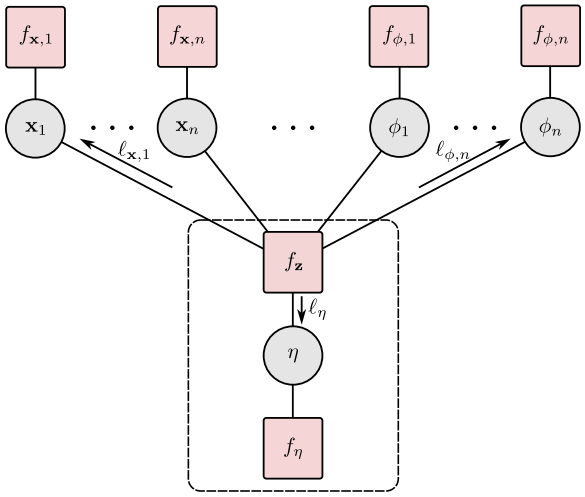


Fig. II.1. Factor graph of the statistical model at one time step for a single measurement block. Circles denote the variable nodes, i.e., the kinematic states $\mathbf{x}_n$, the existence variables and signal powers $\boldsymbol{\phi}_n$, and the noise variance η. Squares denote the factor nodes, i.e., the prior PDFs and the likelihood function $f_{\mathbf{z}}$. Arrows indicate the marginal likelihood messages. The complete factor graph for multiple sensors, dictionaries, and snapshots can be found in [2, Sec. II-E].

TABLE II.1
KEY QUANTITIES FOR THE MODEL (SECTIONS II-A–II-D).

Symbol	Description	Dimension
$\mathbf{z}$	Measurement vector	$\mathbb{C}^C$
$\mathbf{x}_n$	Kinematic state of object n	$\mathbb{R}^d$
$\boldsymbol{\phi}_n = (r_n, \gamma_n)$	Existence variable and signal power	$\{0,1\} \times \mathbb{R}_{>0}$
η	Background noise variance	$\mathbb{R}_{>0}$
$\mathbf{a}(\mathbf{x}_n)$	Array response	$\mathbb{C}^C$
$\bar{\gamma}_n$	Effective mean signal power $f(r_n=1)\,\mathbb{E}[\gamma_n \,\vert\, r_n=1]$	scalar
$\mathbf{R}_n$	Mean outer product of the array response $\mathbb{E}[\mathbf{a}\mathbf{a}^{\mathrm{H}}]$	$C \times C$
$\bar{\eta}$	Mean noise variance $\mathbb{E}[\eta]$	scalar
$\mathbf{C}_{\boldsymbol{\phi},n}$, $\mathbf{C}_{\mathbf{x},n}$, $\mathbf{C}_\eta$	Gaussian PDF covariances	$C \times C$

where we marginalize all other random variables out except for $\boldsymbol{\phi}_n$ itself. Similarly, the marginal likelihood to $\mathbf{x}_n$ is obtained by

$$\ell_{\mathbf{x},n}(\mathbf{x}_n) = \int \cdots \int f(\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta)\, f(\eta)\, d\eta \times \prod_{m=1}^{N} f(\boldsymbol{\phi}_m)\, d\boldsymbol{\phi}_m \prod_{m \neq n} f(\mathbf{x}_m)\, d\mathbf{x}_m. \tag{II.8}$$

The marginal likelihood to noise variable η is obtained by

$$\ell_\eta(\eta) = \int \cdots \int f(\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta) \prod_{m=1}^{N} f(\boldsymbol{\phi}_m)\, d\boldsymbol{\phi}_m \prod_{m=1}^{N} f(\mathbf{x}_m)\, d\mathbf{x}_m. \tag{II.9}$$

C. Gaussian Approximation of Likelihoods

As shown in the derivations from (II.7) to (II.9), the marginal likelihood messages take the form of continuous Gaussian mixtures. We first show that the first moments (mean) of these likelihoods are exactly zero. For example, considering $\ell_{\boldsymbol{\phi},n}$:

$$\begin{aligned}\mathbb{E}_{\ell_{\boldsymbol{\phi},n}}[\mathbf{z}] &= \int \mathbf{z}\, \ell_{\boldsymbol{\phi},n} d\mathbf{z} \\ &= \int \mathbf{z} \left[\int \cdots \int f(\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta) f(\eta) d\eta \prod_{m \neq n} f(\boldsymbol{\phi}_m) d\boldsymbol{\phi}_m \prod_{m=1}^{N} f(\mathbf{x}_m) d\mathbf{x}_m \right] d\mathbf{z} \\ &= \int \cdots \int \underbrace{\left[\int \mathbf{z} f(\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta) d\mathbf{z} \right]}_{= \mathbb{E}[\mathbf{z} \mid \mathbf{x}_{1:N}, \boldsymbol{\phi}_{1:N}, \eta] = \mathbf{0}} f(\eta) d\eta \prod_{m \neq n} f(\boldsymbol{\phi}_m) d\boldsymbol{\phi}_m \prod_{m=1}^{N} f(\mathbf{x}_m) d\mathbf{x}_m \\ &= \mathbf{0},\end{aligned} \tag{II.10}$$

However, a closed-form evaluation of these mixtures is intractable due to the high-dimensional integration over the continuous latent variables. To ensure analytical tractability, we seek to approximate them with simpler distributions. Therefore, we apply second-order moment matching to approximate these likelihoods as single zero-mean Gaussian PDFs:

$$\ell_{\boldsymbol{\phi},n}(\boldsymbol{\phi}_n) \approx \mathcal{CN}\big(\mathbf{z}; \mathbf{0}, \mathbf{C}_{\boldsymbol{\phi},n}\big), \tag{II.11}$$

$$\ell_{\mathbf{x},n}(\mathbf{x}_n) \approx \mathcal{CN}\big(\mathbf{z}; \mathbf{0}, \mathbf{C}_{x,n}\big), \tag{II.12}$$

$$\ell_{\eta}(\eta) \approx \mathcal{CN}\big(\mathbf{z}; \mathbf{0}, \mathbf{C}_{\eta}\big). \tag{II.13}$$

Now we derive how to compute the covariance of each Gaussian in the following sections.

1) Covariance Computation: We now derive the covariance matrices $\mathbf{C}_{\boldsymbol{\phi},n}$, $\mathbf{C}_{x,n}$, and $\mathbf{C}_{\eta}$ in (II.11)–(II.13). Since the outgoing likelihoods are obtained by marginalizing over the *incoming* densities $f_{\mathbf{x},n}$, $f_{\boldsymbol{\phi},n}$, and f_{η}, the required moments are defined with respect to these quantities:

$$\bar{\gamma}_n = \mathbb{E}[r_n \gamma_n] = \sum_{r_n \in \{0,1\}} r_n\, f(r_n) \int \gamma_n\, f(\gamma_n \mid r_n)\, d\gamma_n = f(r_n{=}1)\, \mathbb{E}[\gamma_n \mid r_n{=}1], \tag{II.14}$$

$$\mathbf{R}_n = \mathbb{E}\left[\mathbf{a}(\mathbf{x}_n)\mathbf{a}(\mathbf{x}_n)^{\mathrm{H}}\right] = \int \mathbf{a}(\mathbf{x}_n)\mathbf{a}(\mathbf{x}_n)^{\mathrm{H}}\, f(\mathbf{x}_n)\, d\mathbf{x}_n, \tag{II.15}$$

$$\bar{\eta} = \mathbb{E}[\eta] = \int \eta\, f(\eta)\, d\eta. \tag{II.16}$$

a) Covariance $\mathbf{C}_{\boldsymbol{\phi},n}$: When computing the marginal likelihood to $\boldsymbol{\phi}_n$, the variable $\boldsymbol{\phi}_n$ remains while all other variables are averaged out. Conditioned on all hidden variables, the measurement $\mathbf{z}$ follows a zero-mean complex Gaussian PDF with covariance matrix $\sum_{m=1}^{N} r_m \gamma_m\, \mathbf{a}(\mathbf{x}_m)\mathbf{a}(\mathbf{x}_m)^{\mathrm{H}} + \eta \mathbf{I}_C$. Thus, the second moment of $\mathbf{z}$ is obtained by averaging this conditional covariance over the incoming densities of all variables except $\boldsymbol{\phi}_n$. Hence,

$$\begin{aligned}\mathbf{C}_{\boldsymbol{\phi},n} &= \int \mathbf{z}\mathbf{z}^{\mathrm{H}}\, \ell_{\boldsymbol{\phi},n}\, d\mathbf{z} \\ &= \int \cdots \int \left(\sum_{m=1}^{N} r_m \gamma_m\, \mathbf{a}(\mathbf{x}_m)\mathbf{a}(\mathbf{x}_m)^{\mathrm{H}} + \eta \mathbf{I}_C \right) f(\eta)\, d\eta \prod_{m \neq n} f(\boldsymbol{\phi}_m)\, d\boldsymbol{\phi}_m \prod_{m=1}^{N} f(\mathbf{x}_m)\, d\mathbf{x}_m \\ &= r_n \gamma_n \int \mathbf{a}(\mathbf{x}_n)\mathbf{a}(\mathbf{x}_n)^{\mathrm{H}}\, f(\mathbf{x}_n)\, d\mathbf{x}_n + \left(\int \eta\, f(\eta)\, d\eta \right) \mathbf{I}_C \\ &\quad + \sum_{m \neq n} \left(\int r_m \gamma_m\, f(\boldsymbol{\phi}_m)\, d\boldsymbol{\phi}_m \right) \int \mathbf{a}(\mathbf{x}_m)\mathbf{a}(\mathbf{x}_m)^{\mathrm{H}}\, f(\mathbf{x}_m)\, d\mathbf{x}_m \\ &= r_n \gamma_n\, \mathbf{R}_n + \sum_{m \neq n} \bar{\gamma}_m\, \mathbf{R}_m + \bar{\eta}\, \mathbf{I}_C,\end{aligned} \tag{II.17}$$

where the third and fourth lines follow from the linearity of the integral and the fact that the densities of the variables that do not appear in a term integrate to one, and the last line follows from the definitions of $\bar{\gamma}_m$, $\mathbf{R}_m$, and $\bar{\eta}$ above.

b) Covariance $\mathbf{C}_{\mathbf{x},n}$: When computing the marginal likelihood to $\mathbf{x}_n$, the state $\mathbf{x}_n$ remains while all other variables are averaged out. Thus,

$$\mathbf{C}_{\mathbf{x},n} = \bar{\gamma}_n \, \mathbf{a}(\mathbf{x}_n)\mathbf{a}(\mathbf{x}_n)^{\mathrm{H}} + \sum_{m \neq n} \bar{\gamma}_m \, \mathbf{R}_m + \bar{\eta} \, \mathbf{I}_C. \tag{II.18}$$

Note that the moment matching combines the hypotheses $r_m = 0$ and $r_m = 1$ of each object into the effective mean power $\bar{\gamma}_m = f(r_m = 1)\, \mathbb{E}[\gamma_m \mid r_m = 1]$, and that $\mathbf{C}_{\mathbf{x},n}$ also uses $\bar{\gamma}_n$ for object n itself. Hence, for a potential object with a small existence probability, the message $\ell_{\mathbf{x},n}(\mathbf{x}_n)$ is less informative than the exact Gaussian mixture in (II.8), which can delay the detection of newly appearing objects. This approximation is analogous to the Gaussian moment matching in [2].

c) Covariance $\mathbf{C}_\eta$: When computing the marginal likelihood to η, the noise variable remains while all object contributions are averaged out. Therefore,

$$\mathbf{C}_\eta = \sum_{m=1}^{N} \bar{\gamma}_m \, \mathbf{R}_m + \eta \, \mathbf{I}_C. \tag{II.19}$$

D. Monte Carlo Implementation

The integrals in Section II-C1 involve expectations over continuous priors that generally lack closed-form solutions. We therefore represent each prior by a weighted set of P Monte Carlo samples (particles) and replace integrals with weighted sums.

1) Particle Representation of Prior PDFs:

- **Object state.** The prior of the n-th object state is approximated by a weighted particle set

$$f(\mathbf{x}_n) \approx \left\{ \left(\omega_{\mathbf{x},n,p}, \mathbf{x}_{n,p} \right) \right\}_{p=1}^{P},$$

  where $\mathbf{x}_{n,p}$ denotes the p-th particle and $\omega_{\mathbf{x},n,p}$ is its associated weight, with $\sum_p \omega_{\mathbf{x},n,p} = 1$.
- **Signal power and existence.** The prior of the signal power state $\boldsymbol{\phi}_n = (\gamma_n, r_n)$ is denoted as,

$$f(\boldsymbol{\phi}_n) = \begin{cases} f(r_n = 1)\, f_{\boldsymbol{\theta}}(\gamma_n \mid r_n = 1), & r_n = 1, \\ f(r_n = 0)\, f_{\mathrm{D}}(\gamma_n), & r_n = 0, \end{cases} \tag{II.20}$$

  where $f_{\mathrm{D}}(\gamma_n)$ is a *dummy* density assigned when $r_n = 0$, since the measurement model does not depend on γ_n in this case and it can be chosen arbitrarily without affecting inference. The conditional power prior is approximated by particles

$$f(\gamma_n \mid r_n = 1) \approx \left\{ \left(\omega_{\gamma,n,p}, \gamma_{n,p} \right) \right\}_{p=1}^{P}.$$

  The binary variable r_n follows a Bernoulli distribution, i.e., $f(r_n{=}1) = 1 - f(r_n{=}0)$.
- **Background noise.** The prior of the background noise parameter η is approximated by

$$f(\eta) \approx \left\{ \left(\omega_{\eta,p}, \eta_p \right) \right\}_{p=1}^{P}.$$

In practice, the number of particles used for different random variables need not be the same; here we use the same notation P for simplicity.

2) Per-Particle Covariance Matrices: Substituting the particle representations into the covariance expressions from Section II-C1, each expectation becomes a weighted sum over the corresponding particles. For a given particle of the target variable, the moments of all other variables are computed with respect to their priors, yielding

$$\mathbf{C}_{\boldsymbol{\phi},n,p} = r_n \gamma_{n,p} \, \mathbf{R}_n + \sum_{m \neq n} \bar{\gamma}_m \, \mathbf{R}_m + \bar{\eta} \, \mathbf{I}_C, \tag{II.21}$$

$$\mathbf{C}_{\mathbf{x},n,p} = \bar{\gamma}_n \, \mathbf{a}(\mathbf{x}_{n,p})\mathbf{a}(\mathbf{x}_{n,p})^{\mathrm{H}} + \sum_{m \neq n} \bar{\gamma}_m \, \mathbf{R}_m + \bar{\eta} \, \mathbf{I}_C, \tag{II.22}$$

$$\mathbf{C}_{\eta,p} = \sum_{n=1}^{N} \bar{\gamma}_n \, \mathbf{R}_n + \eta_p \, \mathbf{I}_C, \tag{II.23}$$

where the expected statistics are computed from the prior particles:

$$\bar{\gamma}_n = f(r_n{=}1) \sum_p \omega_{\gamma,n,p}\, \gamma_{n,p}, \tag{II.24}$$

$$\mathbf{R}_n = \sum_p \omega_{\mathbf{x},n,p}\, \mathbf{a}(\mathbf{x}_{n,p})\mathbf{a}(\mathbf{x}_{n,p})^{\mathrm{H}}, \tag{II.25}$$

$$\bar{\eta} = \sum_p \omega_{\eta,p}\, \eta_p. \tag{II.26}$$

3) Weight Update: Each per-particle covariance is used to compute the Gaussian likelihood. The posterior weight of each particle is obtained by multiplying its prior weight with the corresponding likelihood value and renormalizing. This update corresponds to importance sampling with proposal given by the prior PDF, where the importance weights are proportional to the Gaussian likelihood. For example, the weight update for the p-th power particle of object n is

$$\omega^{\mathrm{post}}_{\gamma,n,p} = \frac{\omega_{\gamma,n,p}\, \mathcal{CN}\big(\mathbf{z};\, \mathbf{0},\, \mathbf{C}_{\boldsymbol{\phi},n,p}\big)}{\sum_{p'=1}^{P} \omega_{\gamma,n,p'}\, \mathcal{CN}\big(\mathbf{z};\, \mathbf{0},\, \mathbf{C}_{\boldsymbol{\phi},n,p'}\big)}, \tag{II.27}$$

and similarly for $\omega^{\mathrm{post}}_{\mathbf{x},n,p}$ and $\omega^{\mathrm{post}}_{\eta,p}$. The resulting weighted particle sets form a nonparametric approximation of the marginal posterior PDFs (II.4)–(II.6). After the weight update, a resampling step is applied to mitigate particle degeneracy before propagating the particles of the kinematic states and of the noise variance to the next time step. The prior particles of the signal powers are not propagated; they are drawn anew from the flow at each time step (Section I-C1).

a) Existence probability: The posterior existence probability follows from (II.5) by summing over r_n and integrating over γ_n. For $r_n = 0$, the covariance $\mathbf{C}_{\boldsymbol{\phi},n,p}$ reduces to $\sum_{m\neq n} \bar{\gamma}_m \mathbf{R}_m + \bar{\eta}\, \mathbf{I}_C$, which does not depend on γ_n, and the dummy density $f_{\mathrm{D}}(\gamma_n)$ integrates to one. For $r_n = 1$, the integral over γ_n is replaced by the weighted sum over the power particles. Hence,

$$q(r_n{=}1 \mid \mathbf{z}) = \frac{f(r_n{=}1) \sum_p \omega_{\gamma,n,p}\, \mathcal{CN}\big(\mathbf{z};\mathbf{0},\mathbf{C}_{\boldsymbol{\phi},n,p}\big)}{f(r_n{=}1) \sum_p \omega_{\gamma,n,p}\, \mathcal{CN}\big(\mathbf{z};\mathbf{0},\mathbf{C}_{\boldsymbol{\phi},n,p}\big) + f(r_n{=}0)\, \mathcal{CN}\big(\mathbf{z};\mathbf{0},\sum_{m\neq n} \bar{\gamma}_m \mathbf{R}_m + \bar{\eta}\, \mathbf{I}_C\big)}, \tag{II.28}$$

where $\mathbf{C}_{\boldsymbol{\phi},n,p}$ is evaluated at $r_n = 1$. Moreover, $q(\gamma_n \mid r_n{=}1, \mathbf{z})$ is given by the weights in (II.27), and $q(\gamma_n \mid r_n{=}0, \mathbf{z}) = f_{\mathrm{D}}(\gamma_n)$.

b) Remark: When multiple snapshots, sensors, or dictionaries are available, the measurement consists of several measurement blocks that are conditionally independent given the states, and the factor graph contains one likelihood factor per block. Since the kinematic states and existence variables are shared across blocks, the factor graph contains loops. In this case, the message passing described above is performed for each block, where the prior PDFs of the shared variables are replaced by the products of the prior PDFs and the messages from all other blocks, and the messages are iterated as in [2]. The per-particle weight update rule remains unchanged.

III. Derivations of Efficient Computation

This section derives an efficient computation for the Bayesian inference with high dimensional measurement in our model. All derivations in this section build upon the results of Section II. We proceed in three stages:

1) **Low-rank formulation** (Sections III-A–III-C). A rank-one approximation of the covariance, combined with the Woodbury lemma and the Sherman–Morrison formula, reduces the computation complexity of Gaussian PDFs from $O(C^3PN)$ to $O(CPN^2)$ (Given N objects, P Monte Carlo samples and C-dimensional measurement).
2) **Kronecker Decomposition** (Section III-D). When the manifold admits a Kronecker product structure, all C-dimensional operations could be replaced by operations on the smaller factors, reducing total storage complexity from $O(CPN)$ to $O(C_{\max}NP)$.
3) **Example** (Section III-E). A practical radar sensing example – 3-D tensor analog-to-digital-converter (ADC) data, is used to quantify the specific reductions in computational and storage complexity.

A. Efficient Low-Rank Formulation

Evaluating the Gaussian PDFs directly requires inverting the $C \times C$ covariance matrix at a cost of $O(C^3)$ per evaluation, giving $O(C^3PN)$ over all objects and particles. The storage cost is $O(C^2PN)$. Both are unacceptable when C or P is large. The approach proceeds in three steps:

First, we approximate the covariance of each Gaussian (from (II.7) to (II.9)) by a rank-one outer product (Section III-A1). Second, we exploit this structure via the Woodbury lemma and the Sherman–Morrison formula to compute matrix–vector products involving the inverse without ever forming the $C \times C$ inverse explicitly (Section III-A3). Third, we cast the particle weight updates as log-likelihood ratios (Section III-B), which, combined with the rank-one structure, reduce each likelihood evaluation to two scalars per particle (Section III-C). Table III.1 summarizes the key quantities introduced in Sections III-A–III-C.

TABLE III.1
Key quantities for the low-rank formulation (Sections III-A–III-C).

Symbol	Description	Dimension
$\bar{\mathbf{a}}_n$	Weighted mean array response	$C \times 1$
$\mathbf{u}_n$	$\sqrt{\bar{\gamma}_n}\,\bar{\mathbf{a}}_n$; object contribution	$C \times 1$
$\mathbf{U}$	$[\mathbf{u}_1, \ldots, \mathbf{u}_N]$	$C \times N$
$\mathbf{B}^{-1}$	$(\bar{\eta}\,\mathbf{I}_C + \mathbf{U}\mathbf{U}^{\mathrm{H}})^{-1}$; full covariance	$C \times C$
$\mathbf{G}$	Gram matrix $\mathbf{U}^{\mathrm{H}}\mathbf{U}$	$N \times N$
$\mathbf{L}$	$(\bar{\eta}\,\mathbf{I}_N + \mathbf{G})^{-1}$; inverse kernel	$N \times N$
$\mathbf{A}_n$	$\mathbf{B} - \mathbf{u}_n\mathbf{u}_n^{\mathrm{H}}$; leave-one-out cov.	$C \times C$
$\mathbf{h}_{n,p}$	Particle-specific perturbation	$C \times 1$
$s_{n,p}$, $q_{n,p}$	Log-likelihood ratio scalars	scalar

1) Rank-One Approximation: Recall from Section II-D that $\mathbf{R}_n = \sum_p \omega_{\mathbf{x},n,p}\,\mathbf{a}(\mathbf{x}_{n,p})\mathbf{a}(\mathbf{x}_{n,p})^{\mathrm{H}}$ is a weighted sum of P rank-one outer products, so it has rank up to $\min(P, C)$. Storing or inverting this $C \times C$ matrix is infeasible when C is large. However, in high-dimensional sensing, the measurement likelihood is generally sharply peaked, causing the particle cloud to concentrate as shown in Fig. V.1. This motivates the following approximation:

$$\mathbf{R}_n \approx \left(\sum_p \omega_{\mathbf{x},n,p}\mathbf{a}(\mathbf{x}_{n,p})\right)\left(\sum_p \omega_{\mathbf{x},n,p}\mathbf{a}(\mathbf{x}_{n,p})\right)^{\mathrm{H}} = \bar{\mathbf{a}}_n\bar{\mathbf{a}}_n^{\mathrm{H}}. \tag{III.1}$$

The approximation error equals the matrix $\mathbf{R}_n - \bar{\mathbf{a}}_n\bar{\mathbf{a}}_n^{\mathrm{H}}$, which is positive semidefinite. In practice, because our framework treats the background noise variance η as an unknown random variable rather than a fixed parameter, the Bayesian inference process partly absorbs this residual error. Another way is to manually add a diagonal correction matrix to the background noise

term $\bar{\eta}\mathbf{I}_C$, which will not influence the overall derivations following. The approximation is accurate if the particles of object n are concentrated, so that the phases of the array responses $\mathbf{a}(\mathbf{x}_{n,p})$ are nearly aligned. For a widespread particle cloud, e.g., for a newborn object whose particles are drawn from the birth PDF, these phases average out, and $\|\bar{\mathbf{a}}_n\|^2$ can be much smaller than $\sum_p \omega_{\mathbf{x},n,p}\|\mathbf{a}(\mathbf{x}_{n,p})\|^2$.

2) Notation for the Low-Rank Model: Under the rank-one approximation, we introduce the following quantities:

- $\mathbf{a}_{n,p} \in \mathbb{C}^C$ is the array response of object n for particle p. It is a simplified notation of $\mathbf{a}(\mathbf{x}_{n,p})$ in Section II-D2. The weighted mean is $\bar{\mathbf{a}}_n = \sum_p \omega_{\mathbf{x},n,p}\,\mathbf{a}_{n,p}$.
- $\gamma_{n,p} \in \mathbb{R}$ is the power particle p of object n. The effective mean signal power is $\bar{\gamma}_n = f(r_n{=}1)\sum_p \omega_{\gamma,n,p}\,\gamma_{n,p}$.
- $\eta_p \in \mathbb{R}$ is the noise variance particle p. The weighted mean is $\bar{\eta} = \sum_p \omega_{\eta,p}\,\eta_p$.

We use P to denote the total number of particles, C the dimension of the channel, and N the number of objects. We usually have the following relationships:

$$C \gg N^2 \qquad P \gg N^2$$

The values of dimension C and P depend on the specific problem, but both are typically very large; for the complexity analysis we treat them as being of the same order.

The contribution of object n to the measurement is represented by the C-dimensional vector

$$\mathbf{u}_n = \sqrt{\bar{\gamma}_n} \cdot \bar{\mathbf{a}}_n \in \mathbb{C}^C \tag{III.2}$$

We collect them into a matrix

$$\mathbf{U} = \big[\mathbf{u}_1, \ldots, \mathbf{u}_N\big] \in \mathbb{C}^{C\times N}. \tag{III.3}$$

The covariance of the measurement $\mathbf{z}$ is

$$\mathbf{B} = \bar{\eta}\,\mathbf{I}_C + \mathbf{U}\mathbf{U}^{\mathrm{H}} \in \mathbb{C}^{C\times C}. \tag{III.4}$$

For the object n, we define the covariance when excluding this object:

$$\mathbf{A}_n = \mathbf{B} - \mathbf{u}_n\mathbf{u}_n^{\mathrm{H}}. \tag{III.5}$$

a) The leave-one-out structure: The definition of $\mathbf{A}_n$ comes directly from the sum-product algorithm. Recall from Section II-C1 that the covariance of the likelihood to object n's state or signal power always takes the form

$$\mathbf{C}_{n,p} = \underbrace{\sum_{m\neq n} \bar{\gamma}_m\,\mathbf{R}_m + \bar{\eta}\,\mathbf{I}_C}_{\text{contributions of all other objects + noise}} + \underbrace{\mathbf{h}_{n,p}\mathbf{h}_{n,p}^{\mathrm{H}}}_{\text{particle-specific term of object } n}.$$

That is, when computing the likelihood directed at one variable, the sum-product rule marginalizes out all remaining variables, so object n's own contribution is naturally separated from the rest.

Under the rank-one approximation (III.1), the first part is exactly $\mathbf{A}_n = \mathbf{B} - \mathbf{u}_n\mathbf{u}_n^{\mathrm{H}}$, the covariance of the "background" as seen by object n.

The second part, $\mathbf{h}_{n,p}\mathbf{h}_{n,p}^{\mathrm{H}}$, varies across particles and captures the specific hypothesis about object n in that particle. Because $\mathbf{A}_n$ is shared by all particles of the same object, it only needs to be inverted once per object — an identity that will be used in Section III-B to reduce each likelihood evaluation to two scalars.

3) Woodbury Lemma: The computation of Gaussian PDFs will repeatedly require products of the form $\mathbf{B}^{-1}\mathbf{X}$, where $\mathbf{X}$ varies across objects and particles. By using the Woodbury lemma, the complexity of this matrix multiplication would reduce to $O(CNK)$ where K is the number of columns of $\mathbf{X}$.

For an arbitrary matrix $\mathbf{X} \in \mathbb{C}^{C\times K}$, we have the following.

The inverse of $\mathbf{B}$ is given by the Woodbury lemma

$$\mathbf{B}^{-1} = \frac{1}{\bar{\eta}}\mathbf{I}_C - \frac{1}{\bar{\eta}}\mathbf{U}(\bar{\eta}\mathbf{I}_N + \mathbf{U}^{\mathrm{H}}\mathbf{U})^{-1}\mathbf{U}^{\mathrm{H}}. \tag{III.6}$$

Then

$$\mathbf{B}^{-1}\mathbf{X} = \frac{1}{\bar{\eta}}\mathbf{X} - \frac{1}{\bar{\eta}}\mathbf{U}(\bar{\eta}\mathbf{I}_N + \mathbf{U}^{\mathrm{H}}\mathbf{U})^{-1}\mathbf{U}^{\mathrm{H}}\mathbf{X} \tag{III.7}$$

This is implemented as:

$$\mathbf{L} = (\bar{\eta}\mathbf{I}_N + \mathbf{U}^{\mathrm{H}}\mathbf{U})^{-1}, \tag{III.8}$$

$$\mathbf{T} = \mathbf{L}\mathbf{U}^{\mathrm{H}}\mathbf{X}, \tag{III.9}$$

$$\mathbf{B}^{-1}\mathbf{X} = \frac{1}{\bar{\eta}}(\mathbf{X} - \mathbf{U}\mathbf{T}). \tag{III.10}$$

The two shared quantities $\mathbf{U}^{\mathrm{H}}\mathbf{U} \in \mathbb{C}^{N\times N}$ and $\mathbf{L} = (\bar{\eta}\mathbf{I}_N + \mathbf{U}^{\mathrm{H}}\mathbf{U})^{-1} \in \mathbb{C}^{N\times N}$ are computed once and reused throughout all subsequent computations. Their one-time cost is $O(CN^2)$ for the Gram matrix and $O(N^3)$ for the inversion. After this pre-computation, the cost of each product $\mathbf{B}^{-1}\mathbf{X} \in \mathbb{C}^{C\times K}$ is only $O(CNK)$ if $K > N$.

a) Sherman–Morrison formula: The marginal likelihood for each object n involves the "leave-one-out" covariance $\mathbf{A}_n$, obtained by removing object n's own contribution from $\mathbf{B}$. Rather than recomputing the inverse from scratch for each n, we apply the Sherman–Morrison formula to compute $\mathbf{A}_n^{-1}$. We have

$$\mathbf{A}_n^{-1} = \mathbf{B}^{-1} + \frac{\mathbf{B}^{-1}\mathbf{u}_n\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}}{1 - \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}, \tag{III.11}$$

provided $1 - \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n \neq 0$. This formula can be proved via the Woodbury lemma. The condition always holds, because $\mathbf{A}_n = \bar{\eta}\,\mathbf{I}_C + \sum_{m\neq n}\mathbf{u}_m\mathbf{u}_m^{\mathrm{H}}$ is positive definite and, by the matrix determinant lemma, $1 - \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n = \det(\mathbf{A}_n)/\det(\mathbf{B}) > 0$.

B. Log-Likelihood Ratio

In the Monte Carlo implementation, the weight of each particle is updated by multiplying the prior weight with a likelihood value and renormalizing. Therefore, any factor common to all particles of a given object can cancel out. Hence, it suffices to compute likelihood ratios with respect to a particle-independent reference.

Each particle-specific covariance $\mathbf{C}$ differs from a common factor $\mathbf{A}$ by a single rank-one term: $\mathbf{C} = \mathbf{A} + \mathbf{h}\mathbf{h}^{\mathrm{H}}$. Direct evaluation of a C-dimensional Gaussian PDF requires evaluating $\log\det(\mathbf{C})$ and $\mathbf{z}^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{z}$. Both require $O(C^3)$ operations. However, the *ratio of the likelihood* evaluated with $\mathbf{C}$ to that evaluated with $\mathbf{A}$ can be expressed entirely in terms of **two scalars**. Since $\mathbf{A}$ is the same for all particles, this ratio is all that is needed for weight updates.

The object state and signal-power computation share this common structure, so we derive the log-likelihood ratio once here and apply it repeatedly in Section III-C.

Consider two covariance matrices that differ by a rank-one term:

$$\mathbf{C} = \mathbf{A} + \mathbf{h}\mathbf{h}^{\mathrm{H}}, \tag{III.12}$$

where $\mathbf{A} \in \mathbb{C}^{C\times C}$ is Hermitian positive definite and $\mathbf{h} \in \mathbb{C}^C$ can be any vector. Define the shorthand

$$s = \mathbf{h}^{\mathrm{H}}\mathbf{A}^{-1}\mathbf{h}, \qquad q = \mathbf{z}^{\mathrm{H}}\mathbf{A}^{-1}\mathbf{h}. \tag{III.13}$$

a) Log-determinant: By the matrix determinant lemma,

$$\det(\mathbf{C}) = (1+s)\,\det(\mathbf{A}), \tag{III.14}$$

hence $\log\det(\mathbf{C}) - \log\det(\mathbf{A}) = \log(1+s)$.

b) Quadratic form: By the Sherman–Morrison formula (III.11),

$$\mathbf{C}^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{h}\mathbf{h}^{\mathrm{H}}\mathbf{A}^{-1}}{1+s}, \tag{III.15}$$

so that

$$\mathbf{z}^{\mathrm{H}}\mathbf{A}^{-1}\mathbf{z} - \mathbf{z}^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{z} = \frac{|q|^2}{1+s}. \tag{III.16}$$

c) Log-likelihood ratio: Since the zero-mean complex Gaussian log-PDF is $\log \mathcal{CN}(\mathbf{z}; \mathbf{0}, \mathbf{C}) = -\log\det(\mathbf{C}) - \mathbf{z}^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{z} + \mathrm{const}$, the log-ratio becomes

$$\log \frac{\mathcal{CN}(\mathbf{z}; \mathbf{0}, \mathbf{C})}{\mathcal{CN}(\mathbf{z}; \mathbf{0}, \mathbf{A})} = -\log(1+s) + \frac{|q|^2}{1+s}. \tag{III.17}$$

This identity is applied in the sequel with $\mathbf{A} = \mathbf{A}_n$ (the excluded covariance) and $\mathbf{h} = \mathbf{h}_{n,p}$ (the particle-specific contribution, which will be discussed in detail in the next subsection).

C. Efficient Likelihood Computation

With the log-likelihood ratio identity (III.17) and the efficient inversion tools from Section III-A in hand, we now derive the three types of marginal likelihoods. The key step is to identify the common covariance $\mathbf{A}$ and the rank-one perturbation $\mathbf{h}$, and then evaluate the two scalars s and q from (III.13).

1) Object Update: For each object particle, the particle-specific covariance is $\mathbf{C}_{\mathbf{x},n,p} = \mathbf{A}_n + \mathbf{h}_{n,p}\mathbf{h}_{n,p}^{\mathrm{H}}$, where

$$\mathbf{h}_{n,p} = \sqrt{\bar{\gamma}_n}\, \mathbf{a}_{n,p}. \tag{III.18}$$

Applying (III.17) with $s_{n,p} = \mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{A}_n^{-1}\mathbf{h}_{n,p}$ and $q_{n,p} = \mathbf{z}^{\mathrm{H}}\mathbf{A}_n^{-1}\mathbf{h}_{n,p}$, the log-likelihood message for object n with particle p is

$$\log \ell_{\mathbf{x},n,p} = -\log\big(1+s_{n,p}\big) + \frac{|q_{n,p}|^2}{1+s_{n,p}} + \mathrm{const}. \tag{III.19}$$

a) Formula for $s_{n,p}$: Applying (III.11),

$$\begin{aligned} s_{n,p} = \mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{A}_n^{-1}\mathbf{h}_{n,p} &= \mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} + \mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n\big(1-\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n\big)^{-1}\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} \\ &= \mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} + \frac{\big|\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}\big|^2}{1-\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}. \end{aligned} \tag{III.20}$$

b) Formula for $q_{n,p}$: Likewise, applying (III.11),

$$q_{n,p} = \mathbf{z}^{\mathrm{H}}\mathbf{A}_n^{-1}\mathbf{h}_{n,p} = \mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} + \frac{\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n \cdot \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}}{1-\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}. \tag{III.21}$$

c) Complexity Analysis: Using the pre-computed $\mathbf{U}^{\mathrm{H}}\mathbf{U}$ and $\mathbf{L}$ from Section III-A3, the remaining steps are:

1) Compute $\mathbf{B}^{-1}\mathbf{u}_n$ at cost $O(CN)$ per object; for all N objects: $O(CN^2)$.
2) Compute $\mathbf{B}^{-1}\mathbf{h}_{n,p}$ at cost $O(CN)$ per particle; for all objects and particles: $O(CPN^2)$.

The dominant cost is $O(CPN^2)$.

2) Signal-Power Update: The particle-specific covariance is $\mathbf{C}_{\boldsymbol{\phi},n,p} = \mathbf{A}_n + r_n\, \mathbf{h}_{n,p}\mathbf{h}_{n,p}^{\mathrm{H}}$, where the existence variable r_n appears explicitly because $\boldsymbol{\phi}_n = (r_n, \gamma_n)$ is the message target.

a) Case $r_n = 0$: The rank-one term vanishes, so $\mathbf{C}_{\boldsymbol{\phi},n,p} = \mathbf{A}_n$ and the log-likelihood ratio (III.17) is zero (i.e., $s_{n,p} = q_{n,p} = 0$).

b) Case $r_n = 1$: The particle-specific vector $\mathbf{h}_{n,p}$ is a scaled version of $\mathbf{u}_n$:

$$\alpha_{n,p} = \frac{\sqrt{\gamma_{n,p}}}{\sqrt{\bar{\gamma}_n}}, \qquad \mathbf{h}_{n,p} = \alpha_{n,p}\, \mathbf{u}_n. \tag{III.22}$$

Since $\mathbf{h}_{n,p}$ is proportional to $\mathbf{u}_n$, the quantities in (III.13) simplify to

$$s_{n,p} = \alpha_{n,p}^2\, \mathbf{u}_n^{\mathrm{H}}\mathbf{A}_n^{-1}\mathbf{u}_n, \tag{III.23}$$

$$q_{n,p} = \alpha_{n,p}\, \mathbf{z}^{\mathrm{H}}\mathbf{A}_n^{-1}\mathbf{u}_n. \tag{III.24}$$

Applying (III.17), the log-likelihood message is

$$\log \ell_{\boldsymbol{\phi},n,p} = -\log\big(1+s_{n,p}\big) + \frac{|q_{n,p}|^2}{1+s_{n,p}} + \mathrm{const}. \tag{III.25}$$

c) Formula for $s_{n,p}$: Applying (III.11) and dividing by $\alpha_{n,p}^2$,

$$\frac{s_{n,p}}{\alpha_{n,p}^2} = \mathbf{u}_n^H \mathbf{B}^{-1} \mathbf{u}_n + \frac{\left|\mathbf{u}_n^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n\right|^2}{1 - \mathbf{u}_n^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n}. \tag{III.26}$$

d) Formula for $q_{n,p}$: Likewise, dividing by $\alpha_{n,p}$,

$$\frac{q_{n,p}}{\alpha_{n,p}} = \mathbf{z}^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n + \frac{\mathbf{z}^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n \cdot \mathbf{u}_n^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n}{1 - \mathbf{u}_n^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n}. \tag{III.27}$$

With $\beta_n = \mathbf{u}_n^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n$, these expressions simplify to $s_{n,p}/\alpha_{n,p}^2 = \beta_n/(1-\beta_n)$ and $q_{n,p}/\alpha_{n,p} = \mathbf{z}^{\mathrm{H}} \mathbf{B}^{-1} \mathbf{u}_n/(1-\beta_n)$. To avoid a division by $\sqrt{\bar{\gamma}_n}$, which vanishes if $f(r_n = 1) = 0$, the vector $\mathbf{h}_{n,p} = \sqrt{\gamma_{n,p}}\, \bar{\mathbf{a}}_n$ can also be used directly.

e) Complexity Analysis: The pre-computed quantities from Section III-A3 are reused. The remaining steps are:

1) Compute $\mathbf{B}^{-1}\mathbf{u}_n$ at cost $O(CN)$ per object; for all N objects: $O(CN^2)$.
2) Perform $O(PN)$ scalar multiplications for $\alpha_{n,p}$.

The dominant cost is $O(CN^2)$.

3) Noise Update: Unlike the object and signal-power messages, the noise update does not fit the rank-one ratio framework of (III.17). Instead, we use the eigendecomposition of the Gram matrix $\mathbf{G} = \mathbf{U}^{\mathrm{H}}\mathbf{U}$ to express the log-likelihood entirely in terms of its N eigenvalues. In this way, all C-dimensional matrix operations are reduced to scalar operations.

The covariance we need for each noise particle is

$$\mathbf{C}_p = \eta_p \mathbf{I}_C + \mathbf{U}\mathbf{U}^{\mathrm{H}}. \tag{III.28}$$

Consider the $N \times N$ Gram matrix

$$\mathbf{G} = \mathbf{U}^{\mathrm{H}}\mathbf{U} \in \mathbb{C}^{N \times N}. \tag{III.29}$$

Since $\mathbf{G}$ is Hermitian positive semidefinite, we have the eigen-decomposition

$$\mathbf{G} = \mathbf{V}\, \mathbf{\Lambda}\, \mathbf{V}^{\mathrm{H}}, \qquad \mathbf{V} \in \mathbb{C}^{N \times N} \text{ unitary}, \quad \mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N),\ \lambda_i \geq 0. \tag{III.30}$$

Let $r = \mathrm{rank}(\mathbf{U}) = \mathrm{rank}(\mathbf{G}) \leq N$. Without loss of generality, assume $\lambda_1, \ldots, \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_N = 0$. Partition

$$\mathbf{V} = [\mathbf{V}_r\ \mathbf{V}_0], \qquad \mathbf{\Lambda} = \begin{bmatrix} \mathbf{\Lambda}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix},$$

where $\mathbf{V}_r \in \mathbb{C}^{N \times r}$, $\mathbf{\Lambda}_r = \mathrm{diag}(\lambda_1, \ldots, \lambda_r) \in \mathbb{R}^{r \times r}$.

Define

$$\mathbf{W} = \mathbf{U}\, \mathbf{V}_r\, \mathbf{\Lambda}_r^{-1/2} \in \mathbb{C}^{C \times r}, \tag{III.31}$$

where $\mathbf{\Lambda}_r^{-1/2} = \mathrm{diag}(\lambda_1^{-1/2}, \ldots, \lambda_r^{-1/2})$. It is easy to verify that $\mathbf{W}^{\mathrm{H}}\mathbf{W} = \mathbf{I}_r$. Therefore, we write

$$\mathbf{S} = \mathbf{U}\mathbf{U}^{\mathrm{H}} = \mathbf{W}\, \mathbf{\Lambda}_r\, \mathbf{W}^{\mathrm{H}}. \tag{III.32}$$

Hence $\mathbf{S}$ has at most r nonzero eigenvalues $\lambda_1, \ldots, \lambda_r$ with corresponding eigenvectors given by the columns of $\mathbf{W}$. Using the eigenvalues (recalling that $\mathbf{C}_p = \eta_p \mathbf{I}_C + \mathbf{S}$), we can easily obtain the determinant of $\mathbf{C}_p$

$$\det\big(\mathbf{C}_p\big) = \left(\prod_{i=1}^{r} (\lambda_i + \eta_p)\right) \eta_p^{C-r}, \tag{III.33}$$

hence

$$\log\det\big(\mathbf{C}_p\big) = \sum_{i=1}^{r} \log(\lambda_i + \eta_p) + (C - r)\log\eta_p. \tag{III.34}$$

We use the Woodbury lemma to compute the inverse of $\mathbf{C}_p$,

$$\mathbf{C}_p^{-1} = \frac{1}{\eta_p}\mathbf{I}_C - \frac{1}{\eta_p}\mathbf{W}\,\mathrm{diag}\left(\frac{\lambda_i}{\lambda_i + \eta_p}\right)\mathbf{W}^{\mathrm{H}}. \tag{III.35}$$

Define

$$\begin{aligned}\boldsymbol{\kappa} &= \mathbf{W}^{\mathrm{H}}\mathbf{z} \in \mathbb{C}^r \\ &= \boldsymbol{\Lambda}_r^{-1/2}\mathbf{V}_r^{\mathrm{H}}\mathbf{U}^{\mathrm{H}}\mathbf{z}.\end{aligned}$$

Then

$$\mathbf{z}^{\mathrm{H}}\mathbf{C}_p^{-1}\mathbf{z} = \frac{1}{\eta_p}\|\mathbf{z}\|^2 - \frac{1}{\eta_p}\sum_{i=1}^{r}\frac{|\kappa_i|^2\lambda_i}{\lambda_i+\eta_p}, \tag{III.36}$$

where κ_i is the i-th entry of $\boldsymbol{\kappa}$.

a) Complexity Analysis: The main steps are:

1) Eigen-decomposition of $\mathbf{G}$: $O(N^3)$ (done once).
2) Compute $\boldsymbol{\kappa}$ via $\mathbf{v} = \mathbf{U}^{\mathrm{H}}\mathbf{z}$, and then $\mathbf{s} = \mathbf{V}_r^H\mathbf{v}$, and scaling by $\boldsymbol{\Lambda}_r^{-1/2}$:

$$O(CN) + O(Nr) + O(r) \approx O(CN).$$

3) For each noise particle, evaluate $\log\det(\mathbf{C}_p)$ and $\mathbf{z}^{\mathrm{H}}\mathbf{C}_p^{-1}\mathbf{z}$:

$$O(r) \text{ per particle}, \quad O(Pr) \text{ total}.$$

Thus the total cost is $O(CN + PN)$.

4) Total Complexity: Combining all three messages, the total computational complexity per measurement vector is $O(CPN^2)$. The dominant storage cost comes from the collection $\{\mathbf{h}_{n,p} \in \mathbb{C}^C\}_{n,p}$, giving $O(CPN)$.

D. Kronecker-Structured Array Responses

In many applications the array manifold $\mathbf{a} \in \mathbb{C}^C$ admits a Kronecker product decomposition,

$$\mathbf{a} = \mathbf{a}^{(1)} \otimes \mathbf{a}^{(2)} \otimes \cdots,$$

where each factor $\mathbf{a}^{(i)} \in \mathbb{C}^{C_i}$ ($C = \prod_i C_i$) captures information in a distinct domain. For example, in radar ADC data cubes, the fast-time (range), slow-time (Doppler), and antenna (angle) responses are often approximately separable, although certain waveforms can introduce coupling across dimensions.

The low-rank formulation of the preceding sections already achieves $O(CPN^2)$ computational complexity, but it still requires storing and manipulating full C-dimensional vectors such as $\mathbf{h}_{n,p}$ and $\mathbf{u}_n$. When C is the product of several moderate dimensions—as in radar ADC data cubes where $C = C_{\text{range}} \times C_{\text{Doppler}} \times C_{\text{antenna}}$—the Kronecker structure of the manifold lets us replace every C-dimensional operation with operations on the smaller factors, cutting storage from $O(CPN)$ to $O(C_{\max}NP)$ where $C_{\max} = \max_i C_i$, without increasing the computational order.

This section is organized as follows. (1) After introducing the Kronecker notation and two key identities, we describe the shared pre-computation that replaces all C-dimensional matrix operations with factor-level operations (Section III-D1). (2) We then derive how each of the three message types—object update (Section III-D2), signal-power update (Section III-D3), and noise update (Section III-D4)—is computed entirely in terms of the Kronecker factors, without forming any C-dimensional vector explicitly. Table III.2 summarizes the key quantities introduced in this section.

For simplicity, we consider the two-factor case. For three or more factors, e.g., the fast-time, slow-time, and antenna factors of radar data, the derivation applies analogously: the Gram matrix becomes the Hadamard product of all factor Gram matrices, the inner products with the measurement tensor are computed by successive mode products, and the storage scales with $\sum_i C_i$ instead of $C_{\max}$. The measurement vector $\mathbf{z}$ is the column-wise vectorization of a matrix $\mathbf{Z} \in \mathbb{C}^{C_1 \times C_2}$, i.e., $\mathbf{z} = \mathrm{vec}(\mathbf{Z}) \in \mathbb{C}^C$ with $C = C_1C_2$. Each manifold decomposes as

$$\mathbf{a}_{n,p} = \mathbf{a}_{n,p}^{(1)} \otimes \mathbf{a}_{n,p}^{(2)}, \qquad \mathbf{a}_{n,p}^{(1)} \in \mathbb{C}^{C_2}, \quad \mathbf{a}_{n,p}^{(2)} \in \mathbb{C}^{C_1}, \tag{III.37}$$

TABLE III.2
KEY QUANTITIES FOR THE KRONECKER-STRUCTURED FORMULATION (SECTION III-D).

Symbol	Description	Dimension
$\mathbf{a}_{n,p}^{(i)}$	Kronecker factor of $\mathbf{a}_{n,p}$	$C_i \times 1$
$\mathbf{u}_n^{(i)}$	Kronecker factor of $\mathbf{u}_n$	$C_i \times 1$
$\mathbf{U}^{(i)}$	$[\mathbf{u}_1^{(i)}, \dots, \mathbf{u}_N^{(i)}]$	$C_i \times N$
$\mathbf{G}$	Hadamard of factor Gram matrices	$N \times N$
$\mathbf{v}$	Measurement projection $\mathbf{U}^{\mathrm{H}}\mathbf{z}$	$N \times 1$
$\tilde{\mathbf{v}}$	Measurement projection $\mathbf{L}\mathbf{v}$	$N \times 1$
$\mathbf{y}_{n,p}$	Cross-correlation $\mathbf{U}^{\mathrm{H}}\mathbf{h}_{n,p}$	$N \times 1$

where $\mathbf{a}_{n,p}^{(i)}$ is the i-th factor evaluated at $\mathbf{x}_{n,p}$. Note that, in the particle implementation, the weighted mean $\bar{\mathbf{a}}_n$ is in general not a Kronecker product, it is approximated as $\bar{\mathbf{a}}_n \approx \bar{\mathbf{a}}_n^{(1)} \otimes \bar{\mathbf{a}}_n^{(2)}$ with $\bar{\mathbf{a}}_n^{(i)} = \sum_p \omega_{\mathbf{x},n,p}\, \mathbf{a}_{n,p}^{(i)}$. This approximation is exact if all particles of object n share the same factor in all but one domain and it is accurate for concentrated particle clouds. Correspondingly, $\mathbf{u}_n \approx \mathbf{u}_n^{(1)} \otimes \mathbf{u}_n^{(2)}$, where

$$\mathbf{u}_n^{(1)} = \bar{\gamma}_n^{\frac{1}{4}} \cdot \bar{\mathbf{a}}_n^{(1)},$$
$$\mathbf{u}_n^{(2)} = \bar{\gamma}_n^{\frac{1}{4}} \cdot \bar{\mathbf{a}}_n^{(2)}.$$

Kronecker identities: All subsequent derivations rely on two standard properties of the Kronecker product:

1) For any $\mathbf{a} = \mathbf{a}^{(1)} \otimes \mathbf{a}^{(2)}$,

$$\|\mathbf{a}\|^2 = \|\mathbf{a}^{(1)}\|^2 \, \|\mathbf{a}^{(2)}\|^2. \tag{III.38}$$

2) For $\mathbf{a}^{(1)} \in \mathbb{C}^{C_2}$, $\mathbf{a}^{(2)} \in \mathbb{C}^{C_1}$, and $\mathbf{Z} \in \mathbb{C}^{C_1 \times C_2}$,

$$\mathrm{vec}(\mathbf{Z})^{\mathrm{H}}(\mathbf{a}^{(1)} \otimes \mathbf{a}^{(2)}) = (\mathbf{a}^{(1)})^{\mathrm{T}}\, \mathbf{Z}^{\mathrm{H}}\, \mathbf{a}^{(2)}. \tag{III.39}$$

1) Pre-computation for Shared Terms: We first express the shared building blocks from Sections III-A–III-C in terms of the Kronecker factors. The quantities below are computed once and reused by all three message types.

Gram matrix G: Each entry of $\mathbf{G} = \mathbf{U}^{\mathrm{H}}\mathbf{U} \in \mathbb{C}^{N \times N}$ factorizes by the mixed-product property of the Kronecker product:

$$[\mathbf{G}]_{mn} = (\mathbf{u}_m^{(1)} \otimes \mathbf{u}_m^{(2)})^{\mathrm{H}}(\mathbf{u}_n^{(1)} \otimes \mathbf{u}_n^{(2)}) = (\mathbf{u}_m^{(1)\mathrm{H}}\mathbf{u}_n^{(1)})\,(\mathbf{u}_m^{(2)\mathrm{H}}\mathbf{u}_n^{(2)}), \tag{III.40}$$

which can be written compactly as the Hadamard (element-wise) product of two small Gram matrices:

$$\mathbf{G} = \left(\mathbf{U}^{(1)\mathrm{H}}\mathbf{U}^{(1)}\right) \odot \left(\mathbf{U}^{(2)\mathrm{H}}\mathbf{U}^{(2)}\right). \tag{III.41}$$

Dimension: $N \times N$. Complexity: $O((C_1 + C_2)N^2)$.

Inverse kernel L: We compute the inverse of the regularized Gram matrix once:

$$\mathbf{L} = (\bar{\eta}\mathbf{I}_N + \mathbf{G})^{-1}. \tag{III.42}$$

Dimension: $N \times N$. Complexity: $O(N^3)$.

Measurement projection v and $\tilde{\mathbf{v}}$: Applying (III.39), each entry of $\mathbf{v} = \mathbf{U}^{\mathrm{H}}\mathbf{z} \in \mathbb{C}^N$ can be written as

$$v_n = (\mathbf{u}_n^{(1)\mathrm{H}} \otimes \mathbf{u}_n^{(2)\mathrm{H}})\ \mathrm{vec}(\mathbf{Z}) = (\mathbf{u}_n^{(2)})^{\mathrm{H}}\, \mathbf{Z}\, (\mathbf{u}_n^{(1)})^*, \tag{III.43}$$

Dimension: $N \times 1$. Complexity: $O(CN)$. The filtered projection is then

$$\tilde{\mathbf{v}} = \mathbf{L}\mathbf{v} \in \mathbb{C}^N. \tag{III.44}$$

Dimension: $N \times 1$. Complexity: $O(N^2)$.

Per-object scalar $\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n$: This scalar appears in the Sherman–Morrison expansion for both the object and signal-power updates. Since $\mathbf{u}_n$ is the n-th column of $\mathbf{U}$, the Woodbury lemma yields

$$\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n = \frac{1}{\bar{\eta}}\Big([\mathbf{G}]_{nn} - [\mathbf{GLG}]_{nn}\Big). \tag{III.45}$$

The matrix $\mathbf{GLG}$ is pre-computed once in $O(N^3)$, after which extracting diagonal entries is $O(1)$ per object.

Per-object scalar $\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n$: Similarly, using $\mathbf{v} = \mathbf{U}^{\mathrm{H}}\mathbf{z}$ and $\tilde{\mathbf{v}} = \mathbf{L}\mathbf{v}$,

$$\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n = \frac{1}{\bar{\eta}}\Big(v_n - [\mathbf{G}\tilde{\mathbf{v}}]_n\Big)^*. \tag{III.46}$$

The product $\mathbf{G}\tilde{\mathbf{v}}$ is computed once in $O(N^2)$, making the cost $O(N^2)$ for all N objects combined.

2) Object Update with Kronecker Factors: Recall from Section III-C that the object message for particle p of object n requires the two scalars $s_{n,p}$ and $q_{n,p}$ (III.13), which—via the Sherman–Morrison expansion (III.11)—depend on $\mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}$, $\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}$, and $\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}$. The per-object scalars $\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n$ and $\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n$ are already available from Section III-D1. We now derive the remaining particle-specific terms without forming any C-dimensional vector.

All three terms share the cross-correlation vector

$$\mathbf{y}_{n,p} = \mathbf{U}^{\mathrm{H}}\mathbf{h}_{n,p} = \big(\mathbf{U}^{(1)\mathrm{H}}\mathbf{h}_{n,p}^{(1)}\big) \odot \big(\mathbf{U}^{(2)\mathrm{H}}\mathbf{h}_{n,p}^{(2)}\big), \tag{III.47}$$

by the same Hadamard factorization as (III.41), where $\mathbf{h}_{n,p}^{(1)} = \bar{\gamma}_n^{1/4}\mathbf{a}_{n,p}^{(1)}$. Complexity: $O((C_1{+}C_2)N)$ per particle; $O(C_{\max}N^2P)$ over all objects and particles.

Term $\mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}$: Expanding $\mathbf{B}^{-1}$ via the Woodbury lemma (III.6) and using the norm factorization (III.38) for the first term gives

$$\mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} = \frac{1}{\bar{\eta}}\Big(\|\mathbf{h}_{n,p}^{(1)}\|^2\,\|\mathbf{h}_{n,p}^{(2)}\|^2 - \mathbf{y}_{n,p}^{\mathrm{H}}\mathbf{L}\,\mathbf{y}_{n,p}\Big). \tag{III.48}$$

The first term equals $\|\mathbf{h}_{n,p}\|^2$ but is computed from the C_1- and C_2-dimensional factors, avoiding any C-dimensional operation. Complexity: $O(N^2)$ per particle; $O(N^3P)$ over all objects and particles.

Term $\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}$: Expanding $\mathbf{B}^{-1}$ and applying the vec-Kronecker identity (III.39) to the first inner product,

$$\begin{aligned}\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} &= \frac{1}{\bar{\eta}}\Big(\mathrm{vec}(\mathbf{Z})^{\mathrm{H}}(\mathbf{h}_{n,p}^{(1)} \otimes \mathbf{h}_{n,p}^{(2)}) - \mathbf{v}^{\mathrm{H}}\mathbf{L}\,\mathbf{y}_{n,p}\Big)\\ &= \frac{1}{\bar{\eta}}\Big((\mathbf{h}_{n,p}^{(1)})^{\mathrm{T}}\,\mathbf{Z}^{\mathrm{H}}\,\mathbf{h}_{n,p}^{(2)} - \mathbf{v}^{\mathrm{H}}\mathbf{L}\,\mathbf{y}_{n,p}\Big).\end{aligned} \tag{III.49}$$

The matrix product $\mathbf{Z}^{\mathrm{H}}\,\mathbf{h}_{n,p}^{(2)} \in \mathbb{C}^{C_2}$ costs $O(C_1C_2) = O(C)$, while the second term reuses $\mathbf{L}$ and $\mathbf{y}_{n,p}$. Complexity: $O(C) + O(N^2)$ per particle; $O(CNP)$ over all objects and particles.

Term $\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}$: Noting that $\mathbf{U}^{\mathrm{H}}\mathbf{h}_{n,p} = \mathbf{y}_{n,p}$,

$$\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} = \frac{1}{\bar{\eta}}\Big([\mathbf{y}_{n,p}]_n - [\mathbf{GL}\,\mathbf{y}_{n,p}]_n\Big). \tag{III.50}$$

Complexity: $O(N^2)$ per particle; $O(N^3P)$ over all objects and particles.

Term $s_{n,p}$ and $q_{n,p}$: Substituting the terms above into the Sherman–Morrison expansion from Section III-C, the two scalars that determine the object log-likelihood message (III.19) are

$$s_{n,p} = \mathbf{h}_{n,p}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} + \frac{\big|\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}\big|^2}{1 - \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}, \tag{III.51}$$

$$q_{n,p} = \mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p} + \frac{\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n \cdot \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}_{n,p}}{1 - \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}, \tag{III.52}$$

where every building block on the right-hand side has been expressed in terms of the Kronecker factors. The dominant cost for the object update is reduced to $O(CNP)$.

3) Signal-Power Update with Kronecker Factors: Recall from Section III-C that the signal-power message uses $\mathbf{h}_{n,p} = \alpha_{n,p}\,\mathbf{u}_n$, where $\alpha_{n,p} = \sqrt{\gamma_{n,p}/\bar{\gamma}_n}$. Because $\mathbf{h}_{n,p}$ is proportional to $\mathbf{u}_n$, the scalars $s_{n,p}$ and $q_{n,p}$ reduce to scaled versions of the per-object quantities already computed in Section III-D1:

$$\frac{s_{n,p}}{\alpha_{n,p}^2} = \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n + \frac{\left|\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n\right|^2}{1-\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}, \tag{III.53}$$

$$\frac{q_{n,p}}{\alpha_{n,p}} = \mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n + \frac{\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n \cdot \mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}{1-\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n}. \tag{III.54}$$

Both right-hand sides depend only on $\mathbf{u}_n^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n$ and $\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{u}_n$, which are the per-object scalars from Section III-D1. No additional C- or $C_{\max}$-dimensional operations are needed; the remaining work is $O(PN)$ scalar multiplications.

4) Noise Update with Kronecker Factors: The noise update messages involve the eigendecomposition of the Gram matrix $\mathbf{G}$ and do not depend on the Kronecker structure. The complexity therefore remains $O(CN+PN)$, identical to the non-Kronecker formulation.

5) Total Complexity: Collecting all terms, the total complexity is

- Shared pre-computation (Section III-D1):

$$\underbrace{O(C_{\max}N^2)}_{\mathbf{G}} + \underbrace{O(N^3)}_{\mathbf{L}} + \underbrace{O(CN)}_{\mathbf{v}} + \underbrace{O(N^2)}_{\tilde{\mathbf{v}}} + \underbrace{O(N^3)}_{\text{per-object scalars}} \approx O(CN).$$

- Object update (Section III-D2):

$$\underbrace{O(C_{\max}N^2P)}_{\mathbf{y}} + \underbrace{O(N^3P)}_{\mathbf{h}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}} + \underbrace{O(CNP)}_{\mathbf{z}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}} + \underbrace{O(N^3P)}_{\mathbf{u}^{\mathrm{H}}\mathbf{B}^{-1}\mathbf{h}} \approx O(CNP).$$

- Signal-power update (Section III-D3): $O(PN)$ scalar operations.
- Noise update (Section III-D4): $O(CN+PN)$.

Therefore, the total computational complexity is $O(CPN)$. The key advantage is in storage: only the Kronecker factors $\mathbf{h}_{n,p}^{(1)} \in \mathbb{C}^{C_2}$ and $\mathbf{h}_{n,p}^{(2)} \in \mathbb{C}^{C_1}$ need to be stored, reducing memory from $O(CNP)$ to $O(C_{\max}NP)$ where $C_{\max} = \max(C_1, C_2)$.

E. Example for Complexity Analysis

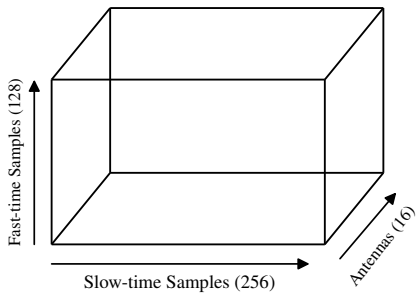


Fig. III.1. Raw ADC data of an FMCW radar arranged as a 3-D tensor with fast-time, slow-time, and antenna dimensions

A typical mmWave radar sensing setup consists of 128 fast-time samples, 256 slow-time samples, and 16 antennas (i.e., $C = 128 \times 256 \times 16$). Consider a 2-D coordinate localization task, where the object state is represented by approximately $P = 10^4$ particles, which is generally sufficient for accurate estimation. Assume there are about $N = 10$ objects in the scene. All computations are performed using complex64/float32 precision. At each time step, a detailed comparison of the three approaches is provided below.

1) Storage:

- **A direct computation of Gaussian PDF**: a single full covariance matrix in $\mathbb{C}^{C\times C}$ requires approximately 2 TiB; storing the covariance matrices of all objects and particles simultaneously would require over 2×10^5 TiB.
- **The low-rank formulation**: the dominant storage is $O(CNP)$, which is around $5 \times 10^{10} \times 8$ bytes ≈ 400 GiB.
- **With Kronecker decomposition**: the dominant storage is reduced to $O((C_1 + C_2 + C_3)NP)$, i.e., $(128 + 256 + 16) \times NP \times 8$ bytes ≈ 0.3 GiB.

For a single likelihood evaluation, i.e., one object and one particle, the full covariance matrix requires approximately 2 TiB, whereas the Kronecker factors of one array response comprise $128 + 256 + 16 = 400$ complex values, i.e., approximately 3.1 KiB; these values are reported in Table I of the main paper.

2) Computation Complexity:

- **A direct computation of Gaussian PDF**: The complexity is $O(C^3NP)$, around 10^{22}.
- **The low-rank formulation**: the complexity is $O(CN^2P)$, around 5×10^{11}.
- **With Kronecker decomposition**: the complexity is $O(CNP)$, around 5×10^{10}.

IV. Additional Results for the Simulated Multisensor Radar Scenario

This section complements Section V-A of the main paper.

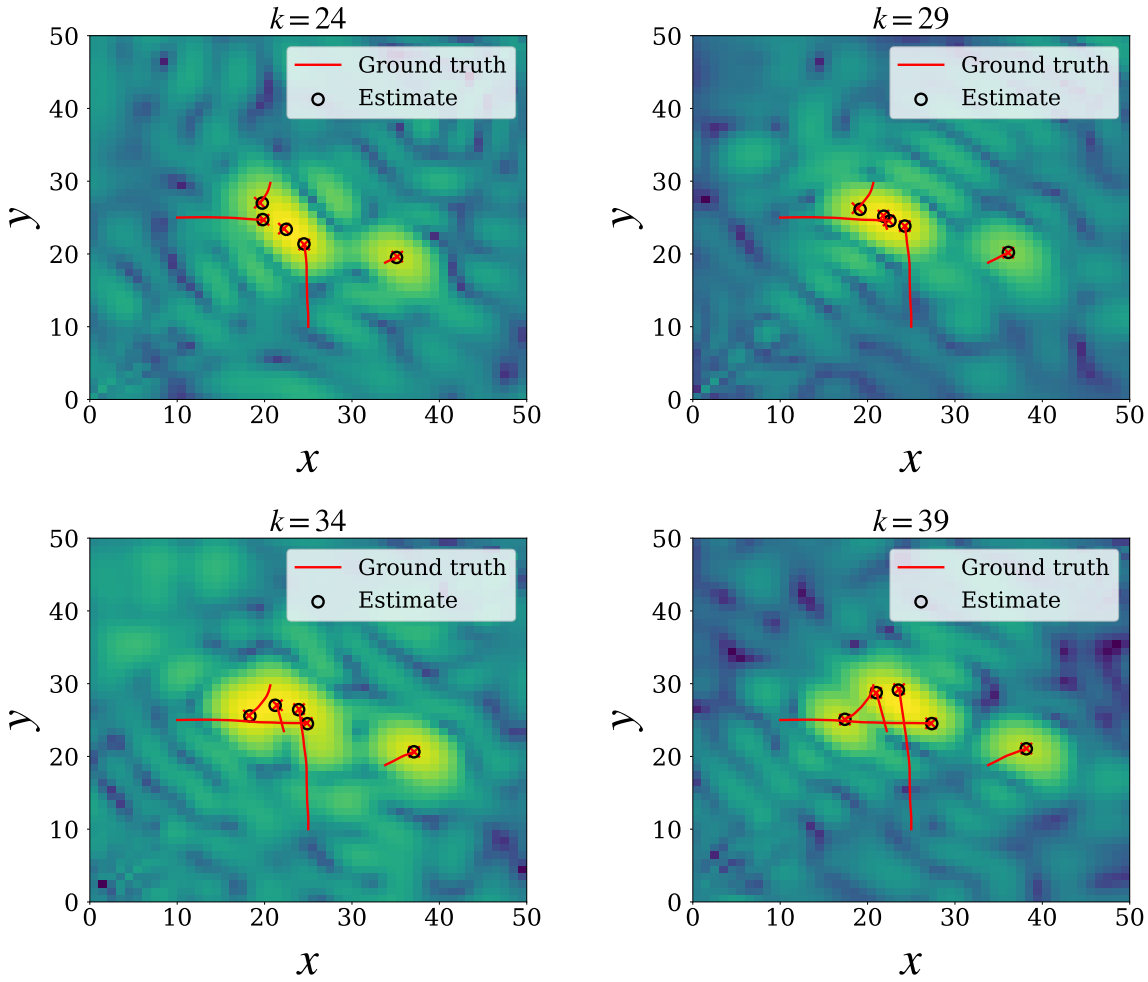


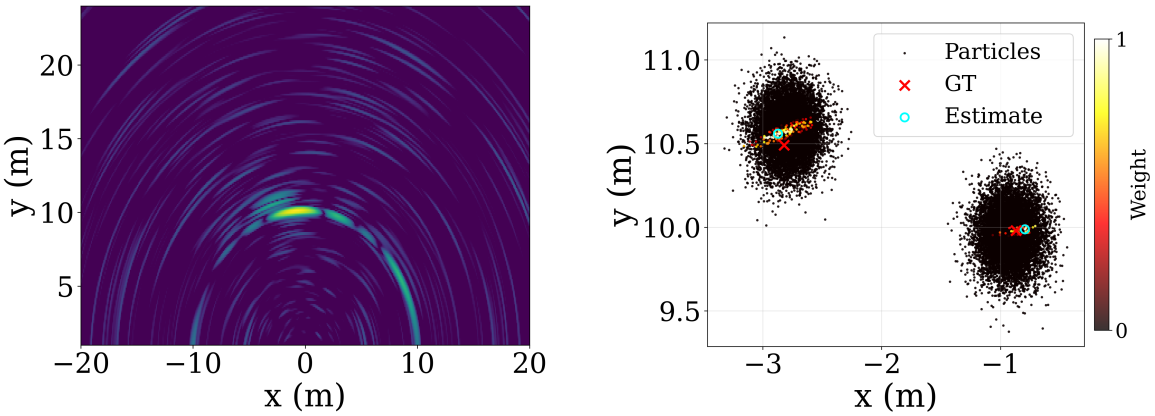
Fig. IV.1. Tracking results obtained by the proposed method in the simulated multisensor radar scenario at time steps $k = 24$, 29, 34, and 39, shown from top left to bottom right. True trajectories are drawn as red curves, true current object positions as red crosses, and estimated object positions as black circles. Matched filtering yields the Bartlett spectrum depicted in the background. Positions are given in meters.

Representative tracking results of the proposed method are depicted in Fig. IV.1. Owing to the low sensor resolution, the responses of closely spaced objects overlap in the Bartlett spectrum obtained by matched filtering, as can be seen, e.g., at $k = 24$. Nevertheless, the proposed method resolves these objects and maintains their tracks.

V. Additional Results for the Active Radio Sensing Experiment

This section complements Section V-B of the main paper.

A. Posterior PDF



(a) Observed power

(b) Posterior PDF

Fig. V.1. (a) Observed spatial power computed from the received complex measurement in a parking-lot scene. (b) Posterior PDF of object positions at the considered time step.

Fig. V.1(a) displays the observed spatial power computed from the received measurement via beamforming, which contains both object-related responses and background variations. Fig. V.1(b) shows a particle representation of the posterior PDF of the object positions at the same time step. The likelihood function of the proposed signal model is highly informative and leads to sharply concentrated posterior PDFs. This concentration motivates the rank-one approximation in Section III-A1.

B. Evolution of the Learned Signal-Power PDF

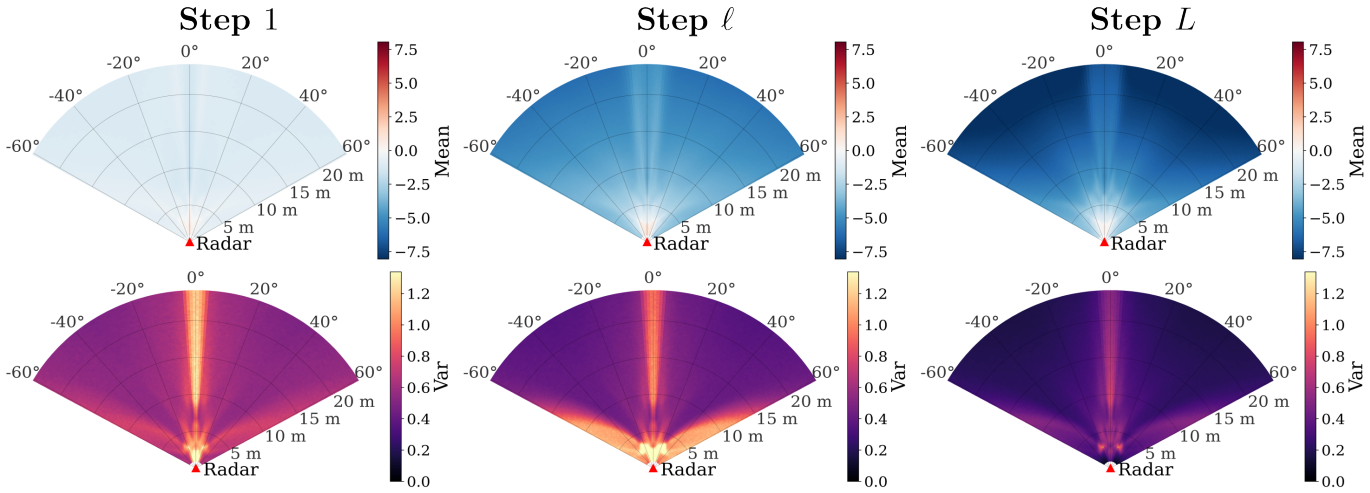


Fig. V.2. Transformation of the radio latent field at intermediate flow steps, with the mean shown in the top row and the variance shown in the bottom row.

Fig. V.2 shows how the learned signal-power PDF evolves through the normalizing flow transformations, using the positional context in (28) of the main paper. At the early stage, the mean field is nearly uniform and the variance is high over most

of the scene. As the samples pass through successive transformations, the mean develops the mainlobe and range-dependent attenuation visible in Fig. 7 of the main paper, while the variance decreases in regions supported by the observations. This evolution provides a distributional view of how the spatial latent field is acquired rather than merely displaying its final mean field.

C. Sensitivity to Birth Initialization

Because the 3-D FFT is used only to initialize new potential objects, we test whether the final performance depends strongly on this coarse proposal. We replace the 3-D FFT peak used as the mean of each Gaussian birth PDF with detections from MVDR or RAMP-CNN and report one run for each strategy in Table V.1. The three strategies yield nearly identical GOSPA

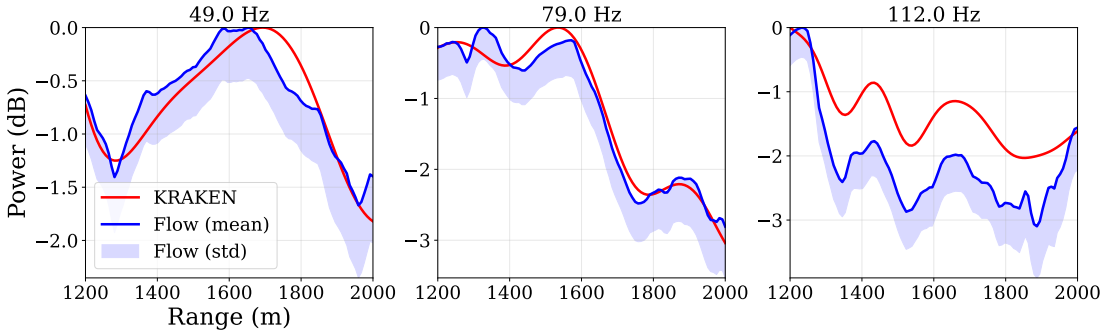
TABLE V.1
GOSPA IN METERS FOR DIFFERENT RADIO BIRTH-INITIALIZATION STRATEGIES, EACH OBTAINED FROM A SINGLE RUN.

Initialization strategy	GOSPA
3-D FFT peak	0.546
MVDR	0.550
RAMP-CNN	0.548

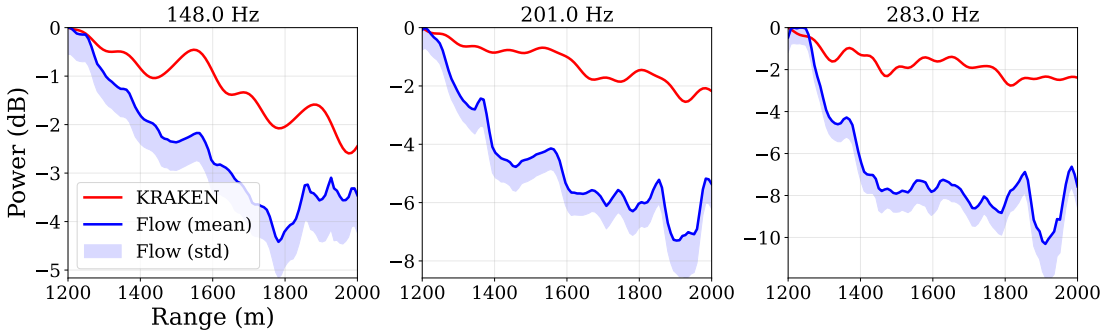
values; since each value is obtained from a single run, the value for the 3-D FFT peak differs slightly from the average over Monte Carlo runs in Table II of the main paper. The birth PDF therefore needs only to identify a plausible search region. Subsequent posterior updates based on the raw measurements dominate the initial point estimate.

VI. Additional Results for the Passive Acoustic Sensing Experiment

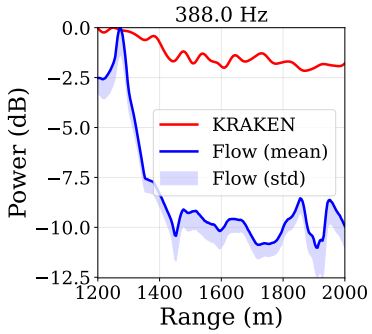
This section complements Section V-C of the main paper. Fig. VI.1 compares the mean field $m_{\boldsymbol{\theta}}(\mathbf{c})$ in (27) of the main paper, i.e., the mean of the learned latent field, with the power predicted by the KRAKEN normal-mode program [4] as a function of range for Event S59 and the fixed estimated source depth of 70 m, for the tones at 49, 79, and 112 Hz in Fig. VI.1a, at 148, 201, and 283 Hz in Fig. VI.1b, and at 388 Hz in Fig. VI.1c; the curves in Fig. VI.1a are also shown in Fig. 9 of the main paper. Each curve is normalized to a maximum of 0 dB, so the comparison concerns the range dependence of the power rather than its absolute level. Possible causes of the growing gap between the curves at high frequencies are discussed in Section V-C of the main paper.



(a) Tones at 49, 79, and 112 Hz



(b) Tones at 148, 201, and 283 Hz



(c) Tone at 388 Hz

Fig. VI.1. Normalized power as a function of range for Event S59 according to the KRAKEN normal-mode program and to the learned latent field at a source depth of 70 m. For the latent field, the mean field $m_{\boldsymbol{\theta}}(\mathbf{c})$ and, for readability, only the lower half of the ± 1 standard-deviation band (shaded) are shown.

VII. Ablation Study of the Signal-Power Prior

This section provides the details of the ablation study summarized in Section V-C5 of the main paper. We perform a controlled acoustic simulation in which KRAKEN generates the measurements and the propagation conditions are known. This experiment compares different spatial and temporal parameterizations while keeping the measurement model, inference procedure, and object trajectories fixed.

a) Controlled environment: For this ablation study, we simulate a shallow-water waveguide with a depth of approximately 216.5 m and a 21-element VLA. The sound-speed profile and the waveguide and seabed parameters used in KRAKEN, as well as the array geometry, are those of the SWellEx-96 environmental model, which is available on the SWellEx-96 website [5]. Between two and six moving sources are distributed over ranges from 500 to 1500 m and depths from 20 to 100 m. Each source transmits the same seven tones from 49 to 388 Hz used in the real-data experiment. The time-varying source number and trajectories produce complex acoustic measurements from which both the object states and latent field must be inferred. The motion and birth models are the same as in the acoustic experiment in Section V-C of the main paper. Each simulation run comprises 60 time steps, and the results are averaged over 100 Monte Carlo runs.

TABLE VII.1

ABLATION STUDY OF THE SIGNAL-POWER PRIOR IN THE CONTROLLED ENVIRONMENT. GOSPA VALUES WITH $c = 100$ M, $\alpha = 2$, AND $p = 1$ ARE GIVEN IN METERS AND AVERAGED OVER 100 MONTE CARLO RUNS.

Method	Temporal	Spatial	GOSPA ↓	Loc. ↓	Miss ↓	False ↓
Flow-G	No	Yes	22.13	12.64	4.47	5.02
Flow-B	Yes	Yes	23.57	13.14	4.78	5.66
Flow-T	Yes	Yes	**21.44**	11.78	4.53	5.12
Gamma	No	Yes	27.16	15.42	6.32	5.42
Gamma-R	Yes	No	31.92	16.27	3.78	11.87

b) Compared latent fields and priors: We compare different normalizing flow designs and conventional parametric alternatives:

1) **Flow-G** uses a stationary Gaussian base and spatial context $\mathbf{c}^{\mathrm{S}}_{k,n}$ containing the estimated source position and tone frequency:
$$\gamma_{k,n} = G\big(T_{\boldsymbol{\theta}}(\varepsilon; \mathbf{c}^{\mathrm{S}}_{k,n})\big), \qquad \varepsilon \sim \mathcal{N}(0,1). \tag{VII.1}$$
2) **Flow-B** introduces temporal information by shifting the base PDF according to the mean $\mu_{q,k-1}$ and variance $\sigma^2_{q,k-1}$ of the preceding posterior of $G^{-1}(\gamma_{k-1,n})$:
$$\varepsilon \sim \mathcal{N}(\mu_{q,k-1}, \sigma^2_{q,k-1}). \tag{VII.2}$$
3) **Flow-T** retains the stationary base $\mathcal{N}(0,1)$ but augments the spatial context $\mathbf{c}^{\mathrm{S}}_{k,n}$ with the mean $\mu_{q,k-1}$ and variance $\sigma^2_{q,k-1}$ of the preceding posterior of $G^{-1}(\gamma_{k-1,n})$, i.e., with the same statistics that Flow-B uses to shift its base PDF. Temporal dependence is therefore represented by the conditional transformation rather than by a time-varying reference PDF.
4) **Gamma** replaces the flow with a context-conditioned Gamma PDF whose parameters are regressed from $\mathbf{c}^{\mathrm{S}}_{k,n}$. The derivations are presented in Section I-D.
5) **Gamma-R** models the power of each source and tone by a Gamma random walk with transition PDF
$$f(\gamma_{k,n} \mid \gamma_{k-1,n}) = \mathrm{Gamma}(\gamma_{k,n};\, \nu,\, \nu/\gamma_{k-1,n}) \tag{VII.3}$$
where the shape $\nu = 100$ is fixed and the rate is obtained by dividing ν by the preceding power. The predicted power therefore has mean $\gamma_{k-1,n}$ and variance $\gamma^2_{k-1,n}/\nu$. This variant has temporal dependence but no explicit spatial context and is representative of a conventional fluctuating-target model [2].

c) Ablation results: Table VII.1 shows that Flow-T achieves the lowest GOSPA and localization error. Compared with Flow-G, adding temporal posterior statistics to the context reduces GOSPA from 22.13 to 21.44. However, temporal information is not uniformly beneficial: Flow-B increases GOSPA to 23.57. Shifting the base PDF creates a time-varying latent reference on top of which the flow must learn a conditional transport map, adding non-stationarity to the ELBO optimization during runtime. Flow-T instead preserves a fixed Gaussian reference and lets the conditional transformation adapt to temporal information, resulting in a more stable and flexible prior.

The parametric alternatives further demonstrate the value of the flow. The Gamma and Gamma-R priors obtain GOSPA values of 27.16 and 31.92, respectively, compared with 21.44 for Flow-T. A single Gamma PDF cannot represent multimodal or heavy-tailed power statistics caused by shallow-water multipath. Gamma-R performs worst overall, with a GOSPA of 31.92 and a particularly large false-object component of 11.87. Its low miss component reflects persistent power hypotheses rather than better overall localization; without spatial context, the random walk cannot follow rapid position-dependent power fluctuations. Together, these results show that both spatial conditioning and a flexible density model are important, and that temporal information is most effective when supplied through the flow context rather than through a moving base PDF.

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